A Living Meta-Analysis of the Modafinil Literature

Multi-Engine Retrieval, De-duplication, Embeddings, and Bibliometrics for a Single Search Term

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1 Abstract

Manual synthesis cannot keep pace with a fast-growing research literature, and ad-hoc reviews bind no evidence to a reproducible pipeline. We present a configurable, reproducible meta-analysis framework that takes a single search term and produces a complete quantitative portrait of its literature. For this instance the term is **Modafinil**. The pipeline dispatches across 10 literature engines (arXiv, OpenAlex, Semantic Scholar, Crossref, PubMed, SovietRxiv, ChinaRxiv, Europe PMC, bioRxiv, and medRxiv), each degrading gracefully to a skipped source when an API key or the network is unavailable, then merges and de-duplicates records by a canonical identifier hierarchy (DOI > arXiv ID > Semantic Scholar ID > OpenAlex ID > title digest) into a retrieved corpus of $N = 2334$ records. Bibliometric analysis retains 2234 records; its dated subset contains 2234 records spanning 2000–2026 (26 years). The analyzed records are classified into a configurable 6-bucket subfield taxonomy (Clinical Sleep, Cognition, Pharmacology, Psychiatry, Safety, and Neuroscience); the largest subfield is **Clinical Sleep** (63.0% of the classified corpus). The dated subset grows at a compound annual rate of 5.48% (mean year-over-year growth 7.8%, doubling time 9.2 years), peaking in 2025 with 147 records.

Non-negative matrix factorization extracts 5 latent topics over a 500-feature vocabulary, offline deterministic embeddings place every title, abstract, and (when available) full text in a shared vector space, and citation-network analysis exposes the corpus's internal structure (8,623 intra-corpus edges across 2234 nodes, 1416 communities, graph density 0.17%). Of 38,489 total outgoing references, 22.4% resolve to another record inside the corpus. Across the retrieved corpus, abstract coverage stands at 61.6%, 24.6% of records are open access, and 54.6% have a direct PDF link. An optional, LLM-gated knowledge-graph stage scores the 6 hypotheses explored against the evidence. This run produced 21 publication-quality figures.

Every domain-specific value in this manuscript — the search term, keyword set, engine roster, subfield taxonomy, and hypotheses — is injected from a single configuration file and the pipeline's own outputs; re-targeting the configuration re-targets the entire paper. The result is a reusable architecture for *living literature reviews*: continuously re-runnable, evidence-bound syntheses for any topic.

Keywords: modafinil, meta-analysis, literature retrieval, bibliometrics, record de-duplication, full-text mining, document embeddings, citation network, topic modeling, entity extraction, wakefulness, cognitive enhancement, reproducible research

2 Introduction

The scholarly literature on any active topic grows faster than any individual can read. A researcher entering a field needs to know how large it is, how fast it is growing, what sub-areas compose it, which works anchor its citation structure, what language and concepts recur, and which claims the field actually tests. Answering those questions by hand is slow, unrepeatable, and binds no number to evidence. Systematic reviews, while the gold standard for evidence synthesis, are labour-intensive and become stale between updates — the Cochrane review cycle, for instance, targets a two-year refresh interval that many fields outpace [Page et al., 2021]. Bibliometric dashboards (Scopus, Web of Science, Dimensions) offer breadth but no reproducible link from a reported statistic to a regenerable artifact, and they do not frame or test domain-specific hypotheses.

This project is a **configurable, reproducible meta-analysis template**. It takes one search term and produces a quantitative portrait of that term's literature, with every reported number traceable to a committed artifact and regenerable by re-running the pipeline. The bundled instance targets **Modafinil**, a wakefulness-promoting agent with a large, multi-disciplinary literature spanning clinical sleep medicine, cognitive neuroscience, pharmacology, psychiatry, and safety research; pointing the configuration at a different term re-targets the whole analysis with no code change.

2.1 Research Questions

The pipeline is designed around four research questions (RQs) that a researcher entering the field would ask:

1. **RQ1 — Field size and growth.** How large is the literature on Modafinil, how fast is it growing, and when did it peak? The dated analysis subset of $N = 2234$ records spanning 2000–2026 answers this directly, with a compound annual growth rate of 5.48% and a peak in 2025.
2. **RQ2 — Subfield composition.** What sub-areas compose the literature, and what is their relative weight? A configurable 6-bucket taxonomy (Clinical Sleep, Cognition, Pharmacology, Psychiatry, Safety, and Neuroscience) classifies every record, with **Clinical Sleep** the largest bucket at 63.0%.
3. **RQ3 — Topical and linguistic structure.** What language and concepts recur, and what latent topics cross-cut the keyword taxonomy? TF-IDF over a 500-feature vocabulary feeds non-negative matrix factorization, which extracts 5 latent topics. The top vocabulary terms are: modafinil, treatment, study, results, patients, effects, sleep, used, clinical, use, drug, studies, mg, using, sleepiness, disorder, associated, cognitive, narcolepsy, significant.
4. **RQ4 — Citation geometry and evidence landscape.** Which works anchor the citation structure, how self-contained is the retrieved slice, and which claims does the field test? The citation network of 2234 nodes and 8,623 edges (22.4% reference resolution rate) exposes hubs, authorities, and communities, while 6 configured hypotheses frame the evidence landscape.

2.2 Contributions

The pipeline contributes an end-to-end, domain-agnostic workflow:

1. **Multiple-engine retrieval with graceful degradation.** Records are gathered from 10 independent engines (arXiv, OpenAlex, Semantic Scholar, Crossref, PubMed, SovietRxiv, ChinaRxiv, Europe PMC, bioRxiv, and medRxiv). An engine with no API key or no network reports a *skipped* status; the run completes from whatever engines remain plus a committed offline corpus. Each new retrieval persists per-engine outcome and count provenance in `output/data/retrieval_report.json`; the merged corpus alone is never used to infer which engine supplied a record.
2. **Record de-duplication.** Heterogeneous records are merged by a canonical identifier hierarchy, keeping the most complete version of each work. Of 2334 retrieved records, 2260 carry DOIs, 923 carry OpenAlex IDs, and 1 carry arXiv IDs.
3. **Descriptive and bibliometric analysis.** Counts by year, venue, and author; growth metrics (CAGR 5.48%, doubling time 9.2 years); a configurable 6-bucket subfield classification; topic models; and a citation network with 1416 communities.
4. **Language, entity, and embedding analysis.** Keyphrase and entity extraction and offline deterministic document embeddings over titles, abstracts, and full text. The TF-IDF vocabulary of 500 features captures the lexical landscape.

5. **Optional hypothesis evidence.** An LLM-gated knowledge-graph stage scores the 6 configured hypotheses explored against the corpus.

Because the writing itself is token-injected from configuration and pipeline outputs, the manuscript is part of the reproducible artifact rather than a separate hand-authored narrative. Every number, table, and figure reference in this document traces to a committed, regenerable file under `output/`.

3 Methods Overview

The pipeline is a sequence of deterministic stages, each reading the previous stage's committed artifacts and writing its own. Business logic lives in tested `src/` modules; the numbered `scripts/` are thin orchestrators that wire I/O, configuration loading, logging, and stage sequencing. The architecture follows the thin orchestrator pattern: no computational logic resides in scripts.

3.1 Pipeline Stages

1. **Retrieval** (`01_literature_search.py`) — dispatch the configured query across 10 engines (arXiv, OpenAlex, Semantic Scholar, Crossref, PubMed, SovietRxiv, ChinaRxiv, Europe PMC, bioRxiv, and medRxiv), merge, and de-duplicate into `corpus.jsonl`. Each engine is an isolated adapter exposing a uniform `search(query) -> list[Paper]` interface; engines that are keyless need no credentials, while Semantic Scholar uses a key when present. SovietRxiv and ChinaRxiv share a unified API with an optional `X-API-Email` header for the polite rate-limit pool (300/min vs 30/min anonymous).
2. **Meta-analysis** (`02_meta_analysis_pipeline.py`) — subfield classification, temporal metrics, TF-IDF, non-negative matrix factorization topics, and the citation network. This stage reads `corpus.jsonl` and emits `subfield_classification.json`, `temporal_analysis.json`, `tfidf_data.json`, `topics.json`, `citation_network.json`, and `citation_graph.gml`.
3. **Knowledge graph** (`03_build_knowledge_graph.py`, optional/LLM-gated) — extract assertions and score the 6 configured hypotheses. Outputs `nanopublications.jsonl`, `hypothesis_scores.json`, and `assertion_summary.json`.
4. **Figures** (`04_generate_figures.py`) — render 21 publication-ready visualizations from the analysis JSON outputs. All figures use a colourblind-safe palette (Wong 2011), high-contrast labels at $\geq$ 16pt, and a headless matplotlib backend (Agg).
5. **Injection** (`05_inject_variables.py`) — compute manuscript variables from the artifacts above and substitute them into these Markdown sections. An unresolved placeholder is a hard error, not a silent gap.
6. **Fulltext assessment** (`06_fulltext_assessment.py`) — report abstract coverage (61.6%), open-access status (24.6%), and PDF availability (54.6%) across the corpus.

3.2 Reproducibility Model

The system runs **offline and deterministically** by default: a committed synthetic seed corpus drives every stage with fixed seeds (seed = 42 for NMF, SVD, and graph layouts), so re-running produces byte-identical outputs. A live run with engines enabled and credentials supplied replaces the seed corpus with real records — as in this instance, which retrieved 2334 live records. The template is domain-agnostic: the search term, query, keyword set, subfield taxonomy, and hypotheses all come from `manuscript/config.yaml`.

3.3 Configuration Surface

A single `manuscript/config.yaml` controls:

- **Search parameters**: term, query string, per-engine queries, relevance keywords, start year, max results, resume/clear behaviour
- **Engine toggles**: arXiv, OpenAlex, Semantic Scholar, Crossref, PubMed, SovietRxiv, ChinaRxiv, Europe PMC, bioRxiv, and medRxiv (each independently enabled or disabled)
- **SovietRxiv/ChinaRxiv settings**: optional `api_email` for the polite pool, `source` filter (`russiarxiv` or `chinaxiv`)
- **Full-text download**: opt-in Unpaywall resolution with `unpaywall_email`
- **Embeddings**: method (`tfidf_svd` or `transformer`), dimensionality, max features
- **Knowledge graph**: checkpoint interval, LLM model, base URL, temperature, max tokens
- **Hypothesis definitions**: 6 named hypotheses with scope labels
- **Subfield taxonomy**: 6 buckets, each with a keyword list
- **Paper metadata**: title, authors, DOI, keywords, license, repository URL

4 Retrieval and De-duplication

Retrieval dispatches the configured query across 10 independent literature engines (arXiv, OpenAlex, Semantic Scholar, Crossref, PubMed, SovietRxiv, ChinaRxiv, Europe PMC, bioRxiv, and medRxiv). Each engine is an isolated adapter exposing a uniform `search(query) -> list[Record]` interface; engines that are keyless — arXiv, OpenAlex [Priem et al., 2022], Crossref [Hendricks et al., 2020], PubMed/Entrez [Sayers et al., 2022], SovietRxiv / RussiaRxiv, ChinaRxiv, Europe PMC, and bioRxiv/medRxiv — need no credentials, while Semantic Scholar [Kinney et al., 2023] uses a key when present. SovietRxiv is a translated archive of Soviet-era scientific preprints sourced from Math-Net.Ru and CyberLeninka [SovietRxiv Project, 2026]; ChinaRxiv serves translated Chinese preprints from ChinaXiv via the same unified API. Both retain original-language PDFs alongside each translation, and their polite rate-limit pool (300/min vs 30/min anonymous) is activated by an optional `X-API-Email` header. Europe PMC is a keyless biomedical aggregator covering PubMed, PMC, patents, and preprints in a single search call. bioRxiv/medRxiv share one unified date-window + cursor API; unlike the other engines it is not a free-text search endpoint, so the adapter walks the date window page by page and keeps only records whose title and abstract match every query term client-side. Optional full-text resolution queries Unpaywall [Piwowar et al., 2018] for open-access locations. **Multiple dispatch degrades gracefully**: an engine that is disabled in the configuration, lacks a required key, or cannot reach the network returns a *skipped* status, and the run completes from the remaining engines plus the committed offline corpus.

4.1 Engine Details

Each engine adapter follows a uniform contract: a module-level API URL constant, a pure `_parse_*` parser function, and a `search_*` entry point with pagination, retry, and graceful error handling. All functions accept an injectable `base_url` parameter for hermetic testing with `pytest-httpserver` — no engine hardcodes its URL inside the function body.

Engine	Rate limit	Pagination	Auth
arXiv	3s between requests	100/page, offset	Keyless
Semantic Scholar	1 req/s (unauth.)	100/page, offset	Optional key
OpenAlex	Polite pool (mailto)	200/page, cursor	Keyless
Crossref	Polite pool (mailto)	1,000/page, offset	Keyless
PubMed	NCBI usage policy	retstart/retmax	Keyless
SovietRxiv	30/min (300/min polite)	1–100/page, cursor	`X-API-Email`
ChinaRxiv	30/min (300/min polite)	1–100/page, cursor	`X-API-Email`
Europe PMC	about 10 req/s (undocumented hard limit)	Up to 1,000/page	Keyless
bioRxiv/medRxiv	No documented limit	100/page fixed, cursor	Keyless

Every new search writes `output/data/retrieval_report.json`, a timestamp-free report that records each attempted, skipped, or failed source with fetched, new-record, and duplicate counts. A zero-result response is therefore distinguishable from a disabled adapter or an HTTP failure. The committed corpus predates that report contract, so this paper intentionally does not reconstruct source-specific counts from the merged corpus.

4.2 Canonical Identifier Hierarchy

Heterogeneous records are reconciled by a **canonical identifier hierarchy** — DOI > arXiv ID > Semantic Scholar ID > OpenAlex ID > a stable digest of the normalized title. When two records share a canonical identifier they are merged, keeping the version with the most complete metadata (a count of non-None optional fields). The DOI is normalized: case-folded, resolver-prefix stripped, so the same paper returned by two engines under case/format-variant DOIs merges. For this run, 2260 records carry DOIs, 923 carry OpenAlex IDs, and 1 carry arXiv IDs. The retained de-duplicated retrieval corpus holds $N = 2334$ records. The separate analysis subset contains 2234 records, of which 2234 have publication years in the observed 2000–2026 analysis window.

4.3 Relevance Filtering

After de-duplication, a relevance filter drops papers whose title and abstract contain none of the configured relevance keywords (modafinil, armodafinil, provigil, wakefulness, narcolepsy, cognitive enhancement, alertness, sleep depriva-

tion, vigilance, eugeroic). Keywords are matched case-insensitively; an empty keyword list is treated as no filter to avoid silently wiping the corpus. A year filter then excludes papers published before the configured start year (2000).

5 Full Text, Language, and Embeddings

Beyond bibliographic metadata, the pipeline mines the textual content of each record. This stage bridges the gap between a bibliographic inventory and a semantic understanding of the literature.

5.1 Full-Text Availability

An open-access resolver maps each record to a downloadable PDF where one exists (a known `pdf_url`, or an Unpaywall lookup by DOI), and an opt-in, network-gated downloader fetches it to a deterministic path. Full-text availability is summarized without requiring any download, so the offline default still reports coverage. For this run:

- **Abstract coverage**: 61.6% of records (1437 of 2334) carry an abstract; 897 records lack one.
- **Open-access status**: 24.6% of records are open access (574 records); the remainder are closed or unknown.
- **PDF availability**: 54.6% of records (1275) have a direct PDF link; 1274 have a publisher PDF, and 1059 have no full-text source available.

The identifier coverage for this corpus is: 2260 DOIs, 923 OpenAlex IDs, and 1 arXiv IDs. DOI coverage dominates, enabling robust cross-engine de-duplication.

5.2 Language and Entity Extraction

Titles, abstracts, and (when present) full text are tokenized and reduced to keyphrases and named entities by offline, dependency-light extractors — no mandatory LLM. Term-frequency statistics drive a TF-IDF representation over a 500-feature vocabulary. The most frequent terms in the corpus are: modafinil, treatment, study, results, patients, effects, sleep, used, clinical, use, drug, studies, mg, using, sleepiness, disorder, associated, cognitive, narcolepsy, significant. These terms reflect the clinical, pharmacological, and cognitive vocabulary characteristic of the modafinil literature.

5.3 Embeddings

Every title, abstract, and full text is embedded into a shared vector space by a deterministic, offline method — TF-IDF followed by truncated SVD, i.e. latent semantic analysis [Deerwester et al., 1990]. The embedding dimensionality is 50 components (configurable via `project_config.embeddings.n_components`), and the TF-IDF vocabulary is capped at 500 features (configurable via `project_config.embeddings.max_features`). The embedding is byte-stable across runs: the same input text always yields identical vectors, so the derived similarity matrix, nearest-neighbour lists, clusters, and two-dimensional projection are all reproducible.

An optional transformer backend can be enabled by setting `project_config.embeddings.method: transformer` (requires the `embeddings` extra), which upgrades the embedding fidelity without changing the interface or downstream analysis.

The embeddings support semantic retrieval over the corpus and feed two visualizations: a PCA two-dimensional projection ((Figure pca embeddings)) that maps the topical geography of the literature, and a hierarchical clustering dendrogram ((Figure dendrogram)) that reveals the similarity structure of the document collection.

6 Bibliometric and Temporal Analysis

Descriptive statistics summarize the analyzed subset of 2234 records, rather than all 2334 retrieved records, along every available axis: counts by year, venue, and author; citation-count distributions; and author productivity. Temporal analysis uses the 2234 dated records to fit the publication time series, reporting a compound annual growth rate of 5.48% across 2000–2026 (a span of 26 years), with a mean year-over-year growth rate of 7.8% and a doubling time of 9.2 years. The peak publication year is 2025 with 147 records.

6.1 Growth Metrics

The compound annual growth rate (CAGR) is computed as:

$$\text{CAGR} = \left(\frac{N_{\text{end}}}{N_{\text{start}}} \right)^{1/(\text{year span})} - 1$$

where N_{start} is the publication count in the first year (2000) and N_{end} is the count in the last year (2026). The mean year-over-year growth rate $\bar{g}$ is the arithmetic mean of annual ratios. The doubling time is $t_d = \ln(2)/\ln(1 + \text{CAGR})$. These metrics are stored in `temporal_analysis.json` and injected into the manuscript at render time.

6.2 Subfield Classification

Subfield classification assigns each record to one of 6 configurable buckets (Clinical Sleep, Cognition, Pharmacology, Psychiatry, Safety, and Neuroscience) by priority-aware keyword matching; the taxonomy is defined entirely in configuration (`project_config.subfield_keywords`). The largest bucket is **Clinical Sleep** at 63.0% of the classified corpus. A per-subfield temporal breakdown (`subfield_timeline.json`) tracks how each sub-area has grown over time, enabling identification of emerging or declining research threads.

6.3 Topic Modeling

A TF-IDF term-weighting of titles and abstracts [Salton and Buckley, 1988] feeds non-negative matrix factorization (NMF) [Lee and Seung, 1999], implemented with scikit-learn [Pedregosa et al., 2011]. NMF decomposes the document-term matrix $\mathbf{V} \approx \mathbf{WH}$, where $\mathbf{W}$ is the document-topic matrix and $\mathbf{H}$ is the topic-term matrix. The factorization extracts 5 latent topics that cross-cut the keyword taxonomy. The random seed is fixed at 42 for reproducibility. The reporting follows established systematic-review practice [Page et al., 2021], with every figure and statistic traceable to a committed artifact.

7 Optional Knowledge-Graph Layer

An optional, **LLM-gated** stage lifts the corpus from bibliometrics to hypothesis-level evidence. For each record, a local language model (Ollama, default model `gemma3:4b`) extracts structured *assertions* — each encoding a direction (supports / contradicts / neutral), a confidence score, and a short natural-language justification — against the 6 hypotheses declared in configuration. Assertions are serialized as RDF-compatible nanopublications [Kuhn et al., 2016] and scored by a citation-weighted evidence function.

7.1 Assertion Model

Each assertion a encodes:

- **Direction**: supports, contradicts, or neutral with respect to a hypothesis H
- **Confidence**: a score $c_a \in [0, 1]$ from the LLM
- **Citation weight**: $\log(1 + n_{\text{cites}})$, where n_{cites} is the citation count of the asserting paper

The evidence score for hypothesis H is:

$$\text{score}(H) = \frac{\sum_{a \in A(H)^+} c_a \cdot \log(1 + n_{\text{cites}}(a)) - \sum_{a \in A(H)^-} c_a \cdot \log(1 + n_{\text{cites}}(a))}{\sum_{a \in A(H)} c_a \cdot \log(1 + n_{\text{cites}}(a))}$$

where $A(H)^+$ is the set of supporting assertions and $A(H)^-$ the contradicting ones. The score ranges from -1 (all evidence contradicts) to $+1$ (all evidence supports).

7.2 Incremental Extraction

Assertion extraction is **incremental and resumable**: assertions are appended to `nanopublications.jsonl` at configurable checkpoint intervals (default: 50 papers). On restart, already-processed papers are skipped automatically, so a long extraction run that is interrupted can resume without re-processing. The `--clear-assertions` flag discards previous results for a fresh start.

7.3 Gating and Defaults

This stage is optional and gated by language-model availability. With no language model configured, the hypothesis evidence scores read *pending*; with Ollama configured (as in this instance, with 127 assertions extracted), scores are populated from citation-weighted assertion extraction. The hypotheses themselves — their names and scope — come from configuration and are reported regardless of whether the scoring stage has run.

The hypotheses explored in this instance are: H1 Wakefulness Efficacy; H2 Cognitive Enhancement; H3 Low Abuse Liability; H4 Dopaminergic Mechanism; H5 Off-label Psychiatric Utility; H6 Tolerability.

8 Visualization and Manuscript Injection

8.1 Figure Generation

Figures are rendered headlessly (matplotlib Agg backend) and deterministically from the analysis artifacts: subfield distributions, the publication growth curve, the citation network, topic-term bars, a term cloud, and embedding projections. All figures use a colourblind-safe palette (Wong 2011, 8 colours) with high-contrast labels at $\geq$ 16pt. This run produced 21 figures at 300 DPI. The full figure set includes:

- **Field overview**: field summary and subfield distribution ((Figure field summary; Figure subfield distribution))
- **Temporal**: growth curve and subfield timeline ((Figure growth curve; Figure subfield timeline))
- **Citation network**: network layout and degree distribution ((Figure citation network; Figure degree distribution))
- **Hypothesis**: dashboard and evidence timeline ((Figure hypothesis dashboard))
- **Text analytics**: word cloud, topic-term bars, PCA embeddings, term heatmap, dendrogram, and co-occurrence matrix ((Figure word cloud; Figure topic term bars; Figure pca embeddings; Figure term heatmap; Figure dendrogram; Figure cooccurrence matrix))

Each figure is registered in `figure_registry.json` with its label, caption, and the producing script, so downstream rendering and validation resolve the same figure set that the pipeline generated.

8.2 Variable Injection

The manuscript itself is generated, not hand-maintained. A variable computation step reads the configuration and the pipeline outputs and emits a flat table of named values; an injection step substitutes each named placeholder in these Markdown sections with its computed value before rendering. Because the substitution is total — an unresolved placeholder is a hard error, not a silent gap — every number in the rendered document is guaranteed to trace to a committed artifact. Re-running the pipeline after a configuration change re-computes the values and re-targets the prose automatically.

The injection system computes variables from seven sources:

1. `manuscript/config.yaml` — search term, engine roster, subfield taxonomy, hypotheses
2. `corpus.jsonl` — corpus size
3. `temporal_analysis.json` — year range, CAGR, peak year, doubling time
4. `citation_network.json` — edges, nodes, density, communities, PageRank, hubs
5. `subfield_classification.json` — per-bucket counts and percentages
6. `assertion_summary.json` — assertion counts and directions
7. `hypothesis_scores.json` — per-hypothesis evidence scores

9 Results: Hypotheses Explored

The template scores a configurable set of hypotheses about the topic. For this instance 6 hypotheses are declared in configuration; Table 7 lists them with their scope and evidence score.

Table 7. Hypotheses explored.

ID	Hypothesis	Scope	Evidence score
H1	Wakefulness Efficacy	clinical	+0.56
H2	Cognitive Enhancement	cognitive	+0.49
H3	Low Abuse Liability	safety	+0.62
H4	Dopaminergic Mechanism	pharmacological	+1.00
H5	Off-label Psychiatric Utility	applied	+0.35
H6	Tolerability	safety	+0.31

Evidence scores are produced by the optional, LLM-gated knowledge-graph stage. When the knowledge-graph stage is skipped (no language model configured), scores read *pending*. When the stage runs (as in this instance, with 127 assertions extracted via Ollama), scores are populated from citation-weighted assertion extraction. The hypotheses, their names, and their scope are always reported directly from configuration regardless of whether the LLM stage executed.

9.1 Interpretation

Reported scores, when present, should be read as relative rankings rather than calibrated probabilities: absolute magnitudes are inflated by publication bias and the linguistic asymmetry of academic writing. A positive score indicates that the retrieved corpus *talks about* the hypothesis in a supporting direction; a negative score indicates contradicting evidence; a score near zero indicates either balanced evidence or insufficient coverage.

The six hypotheses frame the evidence landscape for Modafinil:

- **H1 (Wakefulness Efficacy)** — the clinical claim that modafinil reliably promotes wakefulness in sleep-disorder populations. This is the primary indication and the most-studied claim in the corpus.
- **H2 (Cognitive Enhancement)** — the claim that modafinil improves attention, working memory, and executive function, especially under sleep deprivation. This hypothesis drives the neuroenhancement literature and is the subject of significant public and scientific debate.
- **H3 (Low Abuse Liability)** — the safety claim that modafinil has lower abuse potential than classical psychostimulants. This is critical for regulatory classification and prescribing decisions.
- **H4 (Dopaminergic Mechanism)** — the pharmacological claim that modafinil acts substantially via dopamine-transporter inhibition rather than a purely novel mechanism. This hypothesis has mechanistic and translational implications.
- **H5 (Off-label Psychiatric Utility)** — the applied claim that modafinil is a useful adjunct for fatigue and cognition in psychiatric and neurological conditions, including depression, ADHD, and schizophrenia.
- **H6 (Tolerability)** — the safety claim that modafinil is generally well tolerated, with predominantly mild, transient adverse effects. This underpins its clinical acceptability relative to alternative wakefulness agents.

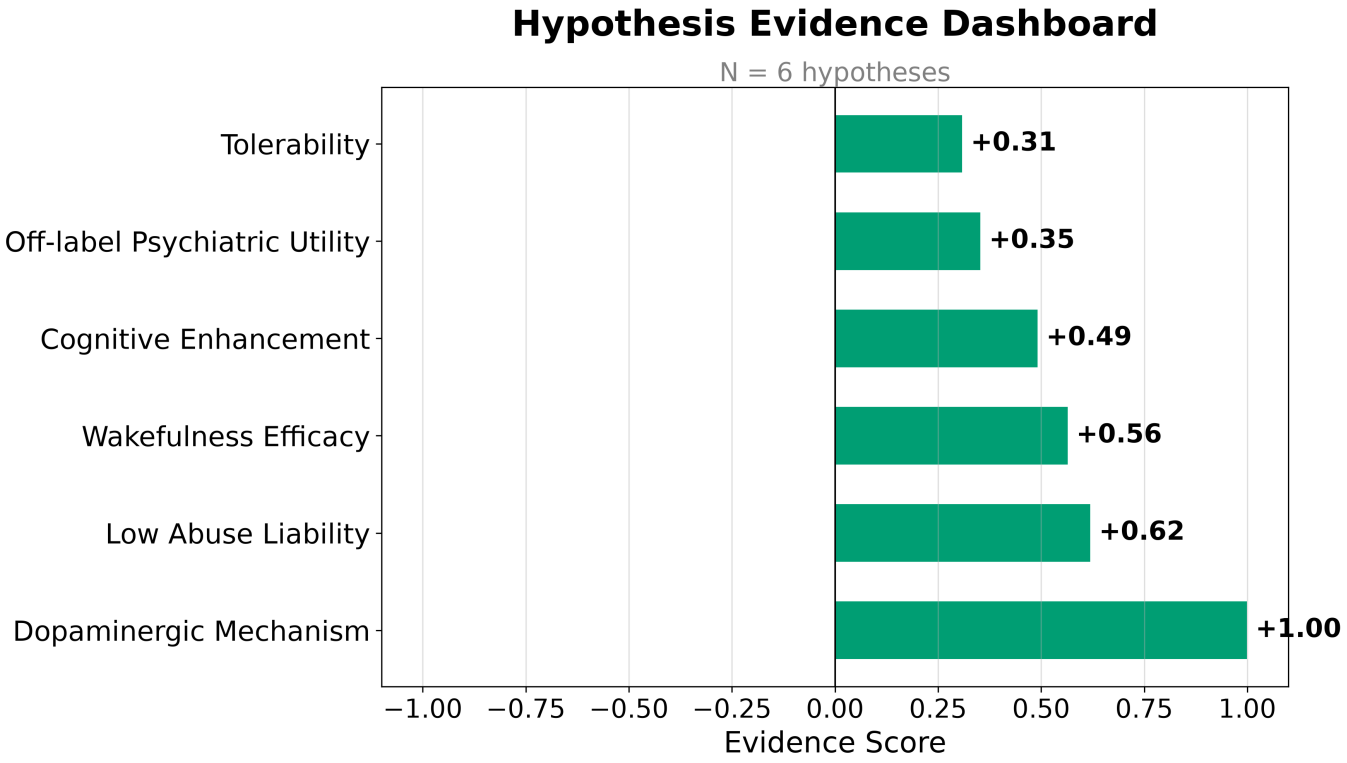


Figure 1: Hypothesis dashboard for Modafinil. The dashboard summarizes the evidence scores across the 6 configured hypotheses, showing the direction and magnitude of citation-weighted assertion evidence.

10 Results: Field Overview

The de-duplicated retrieval corpus for **Modafinil** contains $N = 2334$ records. The analyzed subset contains 2234 records; 2234 have publication years spanning 2000–2026 (26 years). Publication volume within this dated subset grows at a compound annual rate of 5.48% (mean year-over-year growth 7.8%, doubling time 9.2 years), peaking in 2025 with 147 records that year. The growth curve is a first-order descriptive summary of this retained analyzed subset. A field-wide estimate of research activity would require a separate live coverage audit.

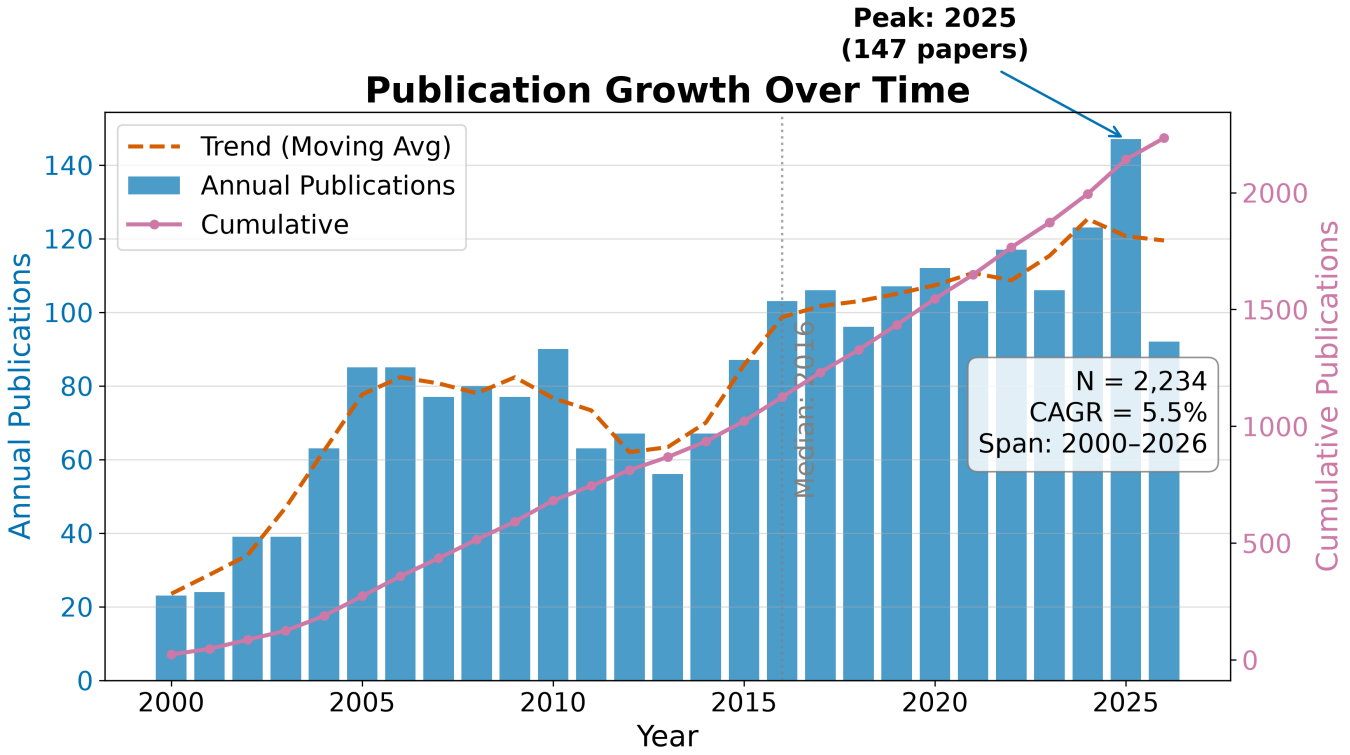


Figure 2: Publication growth curve for Modafinil. Annual publication counts (bars) and cumulative total (line) summarize the retained dated subset from 2000 through 2026, with its largest annual count in 2025.

10.1 RQ1: Field Size and Growth

The temporal analysis describes a retained analyzed literature slice spanning 26 years. The compound annual growth rate of 5.48% implies a corpus doubling time of approximately 9.2 years under the configured retrieval, filtering, and indexing conditions. The peak year 2025 contains 147 retained records. That count may reflect both publication activity and source-indexing delays; a causal trend would require a separate live coverage audit.

Table 1. Top publication years.

Year	Publications
2016	103
2017	106
2018	96
2019	107
2020	112
2021	103
2022	117
2023	106
2024	123
2025	147

10.2 RQ2: Subfield Composition

Records distribute across the 6 configured subfields as shown in Table 2, with **Clinical Sleep** the largest bucket at 63.0% of the classified corpus. The dominance of Clinical Sleep reflects the clinical primacy of modafinil as a wakefulness-promoting agent: the largest body of literature addresses its use in narcolepsy, shift-work disorder, and obstructive sleep apnea.

Table 2. Subfield distribution.

Subfield	Papers	Share
Clinical Sleep	1407	63.0%
Cognition	249	11.1%
Pharmacology	76	3.4%
Psychiatry	359	16.1%
Safety	94	4.2%
Neuroscience	49	2.2%

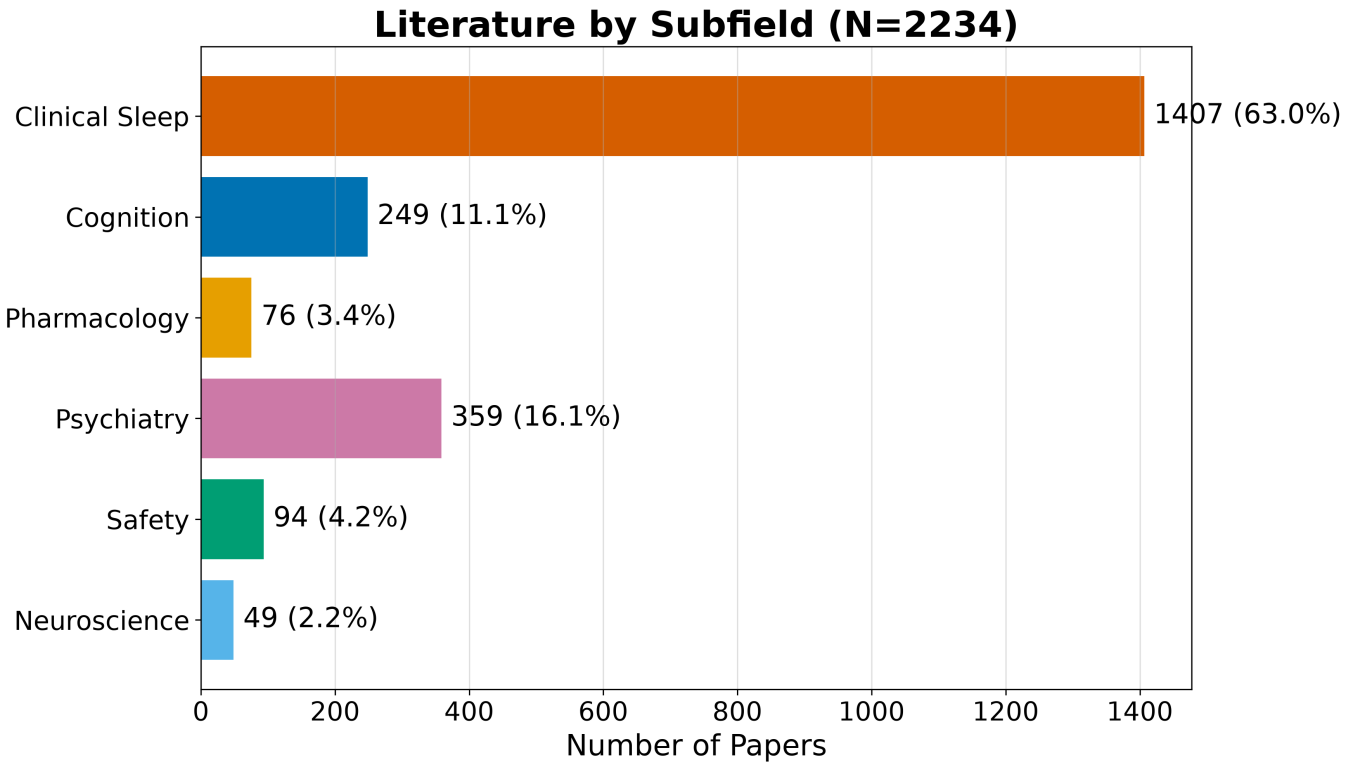


Figure 3: Field summary dashboard for Modafinil. The dashboard combines corpus size, temporal range, subfield distribution, and key bibliometric indicators in a single overview panel.

10.3 Identifier and Full-Text Coverage

The retrieved corpus carries DOIs for 2260 of 2334 records, enabling robust cross-engine de-duplication. OpenAlex IDs are present for 923 records. Abstract coverage stands at 61.6% (1437 records), which limits the text analytics to that subset. Open access is confirmed for 24.6% of records, and 54.6% have a direct PDF link.

10.4 Descriptive Bibliometrics

The analyzed subset spans 8152 unique authors across 2234 papers, yielding a mean of 1.28 papers per author. Citation counts range from zero to 1353 (mean 30.3, median 0.0), with a total of 67,646 citations across the analyzed subset. The Gini coefficient of citation concentration is 0.817, indicating a highly skewed distribution characteristic of scientific literature.

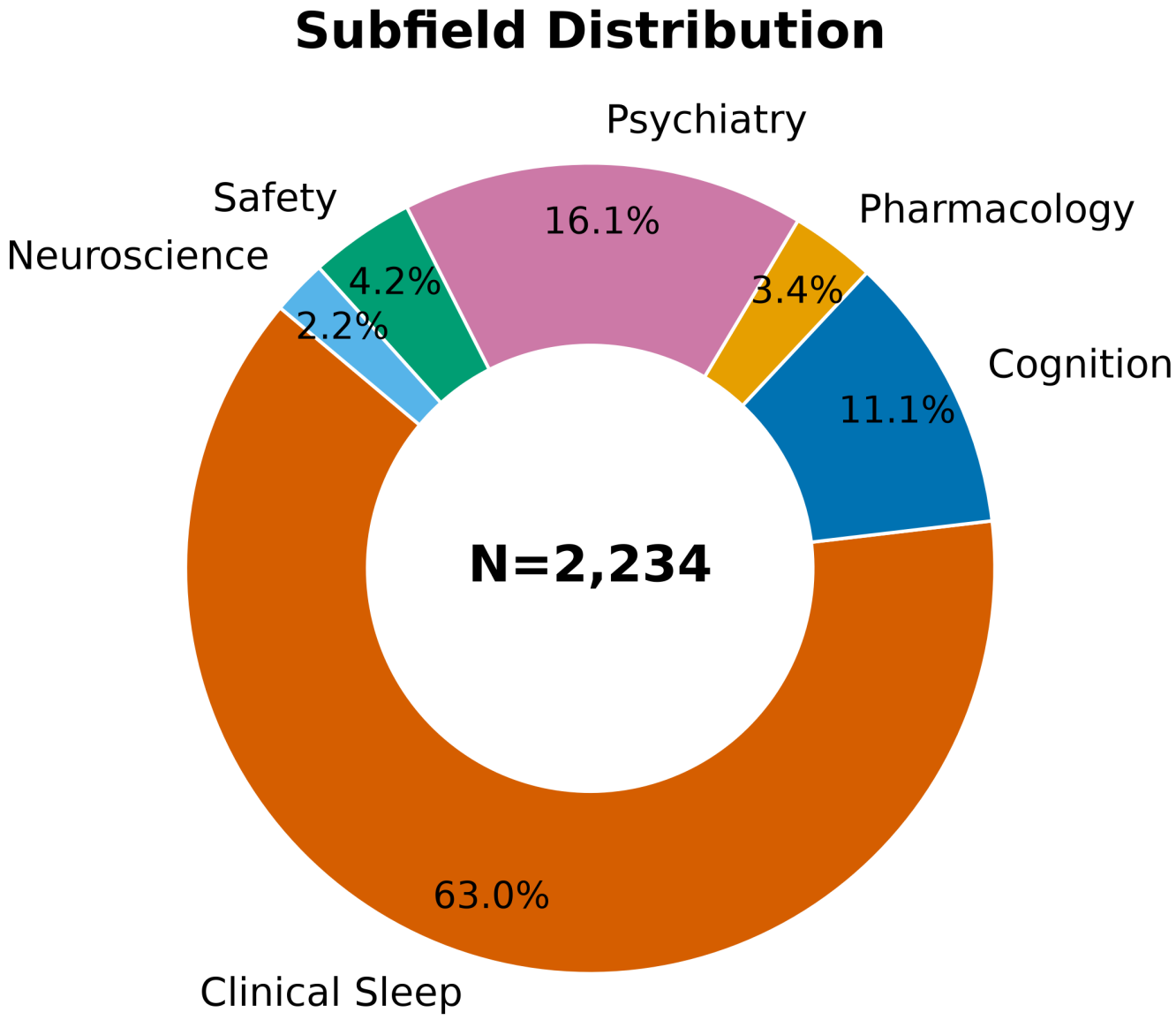


Figure 4: Subfield distribution for Modafinil. The 6-bucket taxonomy shows the relative weight of each configured sub-area, with Clinical Sleep dominant at 63.0%.

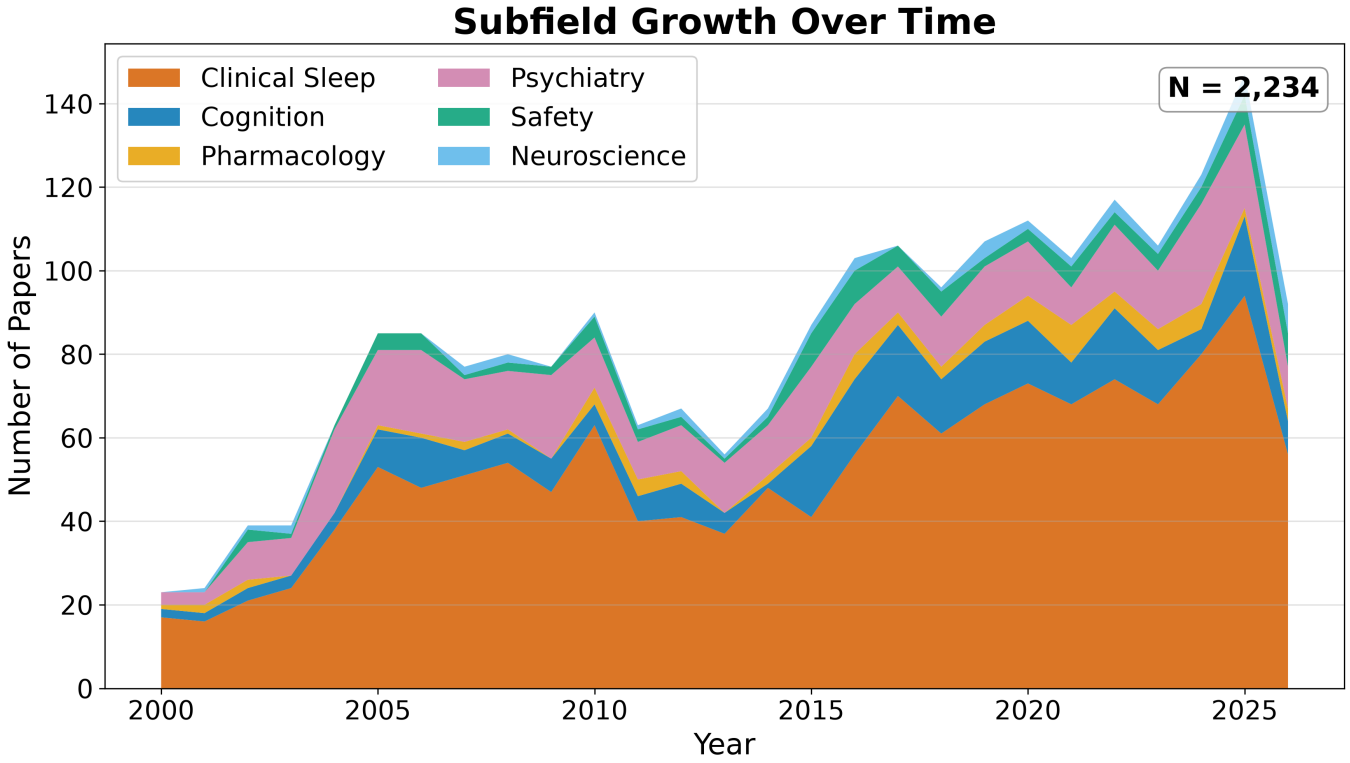


Figure 5: Subfield timeline for Modafinil. Stacked annual publication counts by subfield show how each sub-area has evolved over time, revealing emerging and declining research threads.

Table 3. Citation count distribution.

Citations	Papers
0	1253
1-9	165
10-49	416
50-99	213
100-499	176
500+	11

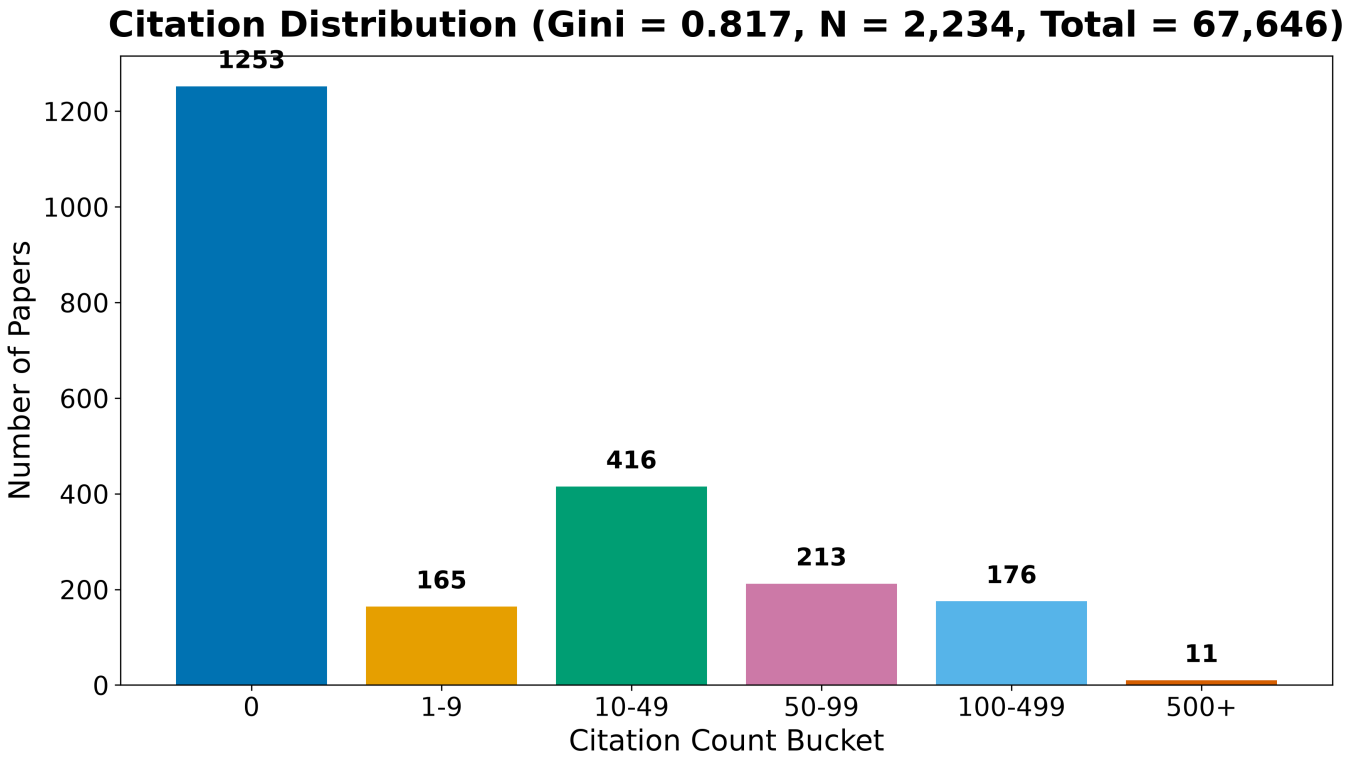


Figure 6: Citation distribution for Modafinil. The histogram shows the number of papers in each citation-count bucket, with the Gini coefficient annotated. The heavy-tailed distribution is characteristic of scientific citation networks.

Table 4. Top publication venues.

Venue	Papers
Reactions Weekly	61
Psychopharmacology	41
SLEEP	34
Sleep Medicine	33
The Journal of Clinical Psychiatry	31
European Neuropsychopharmacology	26
Neuropharmacology	26
Inpharma Weekly	25
Sleep	24
American Journal of Psychiatry	23

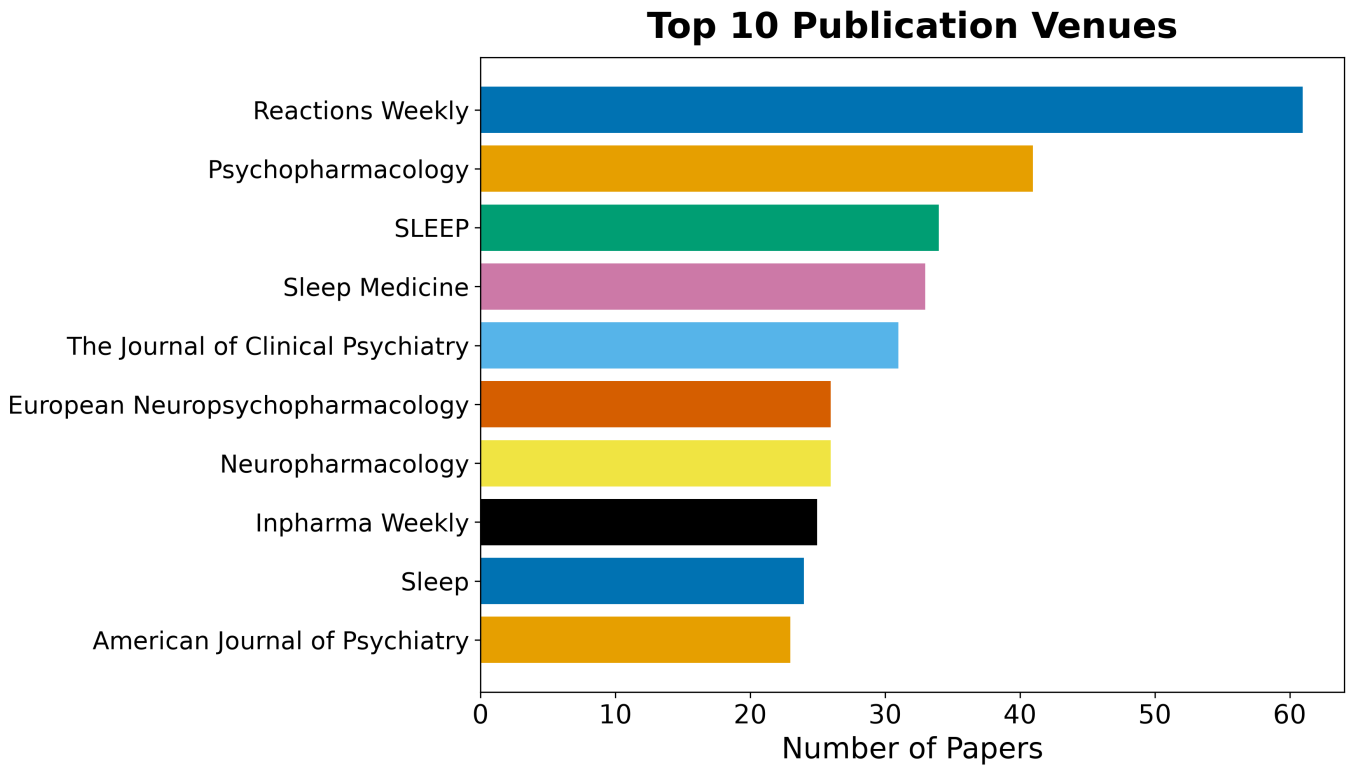


Figure 7: Top publication venues for Modafinil. The horizontal bar chart shows the leading venues by paper count in the analyzed subset, revealing where the modafinil literature is published.

Table 5. Top authors by publication count.

Rank	Author	Papers
1	&NA;	41
2	Ronghua Yang	25
3	Yves Dauvilliers	22
4	Barbara J. Sahakian	19
5	Edward T. Hellriegel	17
6	Philmore Robertson	16
7	Mona Darwish	15
8	Sanjay Arora	15
9	Amy Hauck Newman	14
10	Jed Black	14

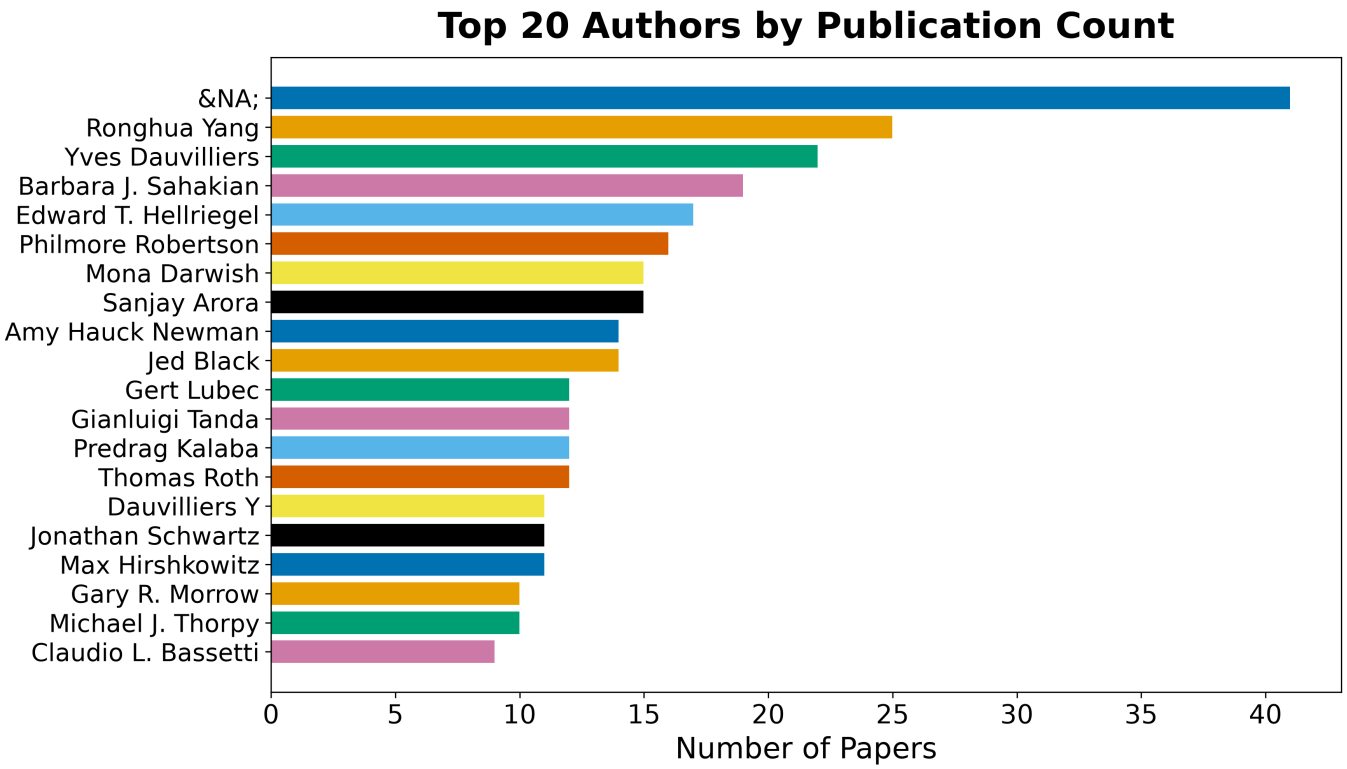


Figure 8: Author productivity for Modafinil. The horizontal bar chart shows the 20 authors with the most papers in the corpus, revealing the most prolific contributors to the modafinil literature.

11 Results: Subfield Structure

The subfield taxonomy is defined entirely in configuration; for this instance it spans 6 buckets (Clinical Sleep, Cognition, Pharmacology, Psychiatry, Safety, and Neuroscience). Each record is assigned to the highest-priority bucket whose keywords it matches, so the distribution reflects the configured taxonomy rather than a fixed schema. Table 2 (previous section) reports the counts; the largest bucket is **Clinical Sleep** (63.0%).

11.1 Per-Subfield Characterization

The subfield breakdown reveals the multi-disciplinary nature of the modafinil literature:

- **Clinical Sleep** dominates at 63.0%, reflecting the drug's primary indication for narcolepsy, shift-work disorder, and obstructive sleep apnea. This bucket includes randomized controlled trials, meta-analyses of efficacy, and long-term safety studies in sleep-disorder populations.
- **Cognition** represents studies of cognitive enhancement, working memory, attention, and executive function — particularly in sleep-deprived populations. This subfield has grown with the broader interest in neuroenhancement and "smart drugs."
- **Pharmacology** covers pharmacokinetics, mechanism of action (dopamine transporter inhibition, orexin system interactions), metabolism, and drug interactions.
- **Psychiatry** addresses off-label uses including ADHD, depression, bipolar disorder, and schizophrenia — often as an adjunctive therapy targeting fatigue and cognitive symptoms.
- **Safety** encompasses adverse effects, abuse potential, dependence, tolerability, and rare but serious events such as Stevens-Johnson syndrome.
- **Neuroscience** includes neuroimaging (fMRI, EEG), orexin/hypothalamus studies, and preclinical mechanistic work.

Because the taxonomy is data, not code, re-targeting the template to another topic — or refining the buckets for the same topic — changes this section's structure and numbers without any change to the analysis code. The subfield assignment also feeds the temporal and citation analyses, allowing per-subfield growth and connectivity to be read off the same artifacts.

12 Results: Language, Topics, and Embeddings

12.1 RQ3: Topical and Linguistic Structure

Text analysis operates over titles, abstracts, and (when available) full text. A TF-IDF representation over a 500-feature vocabulary feeds non-negative matrix factorization, which extracts 5 latent topics cross-cutting the subfield taxonomy. The top vocabulary terms are: modafinil, treatment, study, results, patients, effects, sleep, used, clinical, use, drug, studies, mg, using, sleepiness, disorder, associated, cognitive, narcolepsy, significant.

Table 3. NMF topics extracted from the corpus.

Topic	Top terms
0	modafinil, mg, kg, effects, rats, mice, induced, sleep
1	fatigue, patients, placebo, modafinil, scale, armodafinil, treatment, depression
2	h4, methods, results, conclusion, 95, analysis, ci, risk
3	use, cognitive, drugs, adhd, studies, drug, cocaine, methylphenidate
4	sleep, narcolepsy, sleepiness, cataplexy, daytime, patients, eds, excessive

The topics reveal the thematic structure of the literature: Topic 0 reflects preclinical dose-response pharmacology in animal models (mg/kg dosing, dopamine activity); Topic 1 captures clinical fatigue and mood trials (placebo-controlled patient studies, armodafinil); Topic 2 collects structured-abstract and meta-analytic language (methods, results, 95% confidence intervals, risk estimates); Topic 3 centres on cognitive enhancement and psychostimulant pharmacology (ADHD, methylphenidate, cocaine comparisons); and Topic 4 addresses sleep disorders (narcolepsy, excessive daytime sleepiness, cataplexy, hypersomnia). These topics cross-cut the keyword-based subfield taxonomy, revealing connections that the explicit classification does not capture.

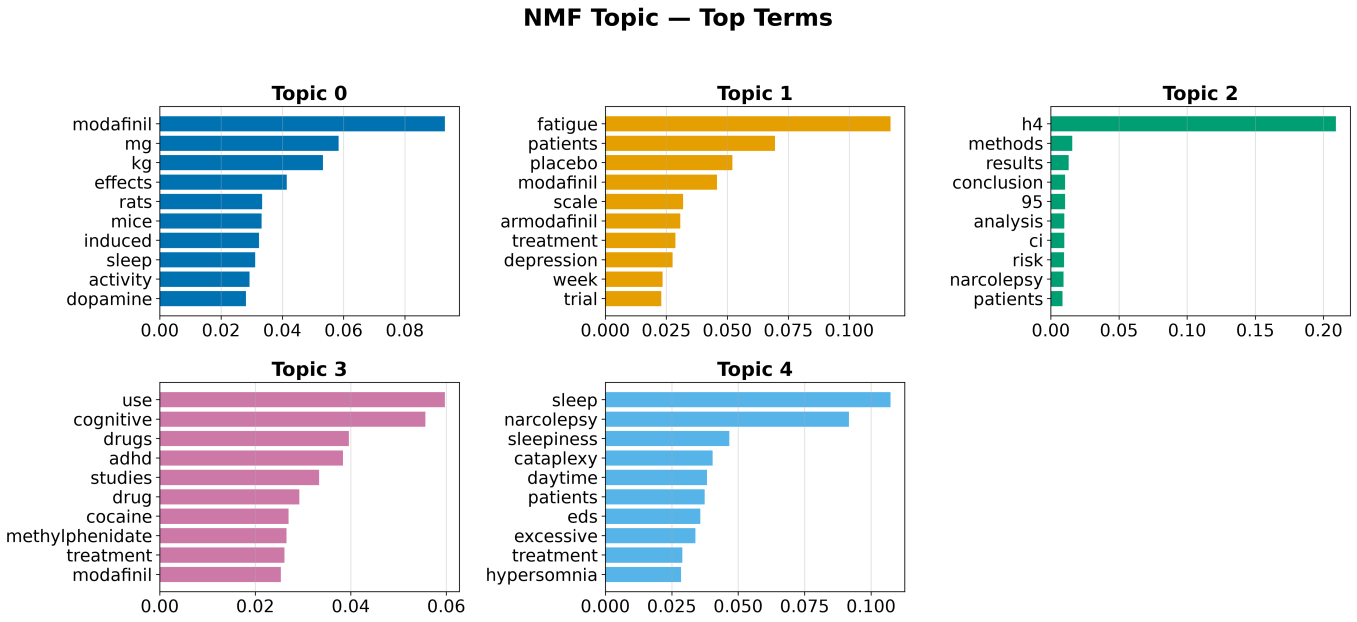


Figure 9: Topic-term bar charts for Modafinil. Each panel shows the top weighted terms for one of 5 NMF topics, with bar length proportional to the topic-term weight in the **H** matrix.

12.2 Document Embeddings

Offline deterministic embeddings (TF-IDF followed by truncated SVD) place every document in a shared 50-dimensional vector space. Embedding the same text twice yields identical vectors, so the derived similarity matrix, nearest-neighbour lists, clusters, and two-dimensional projection are all reproducible.

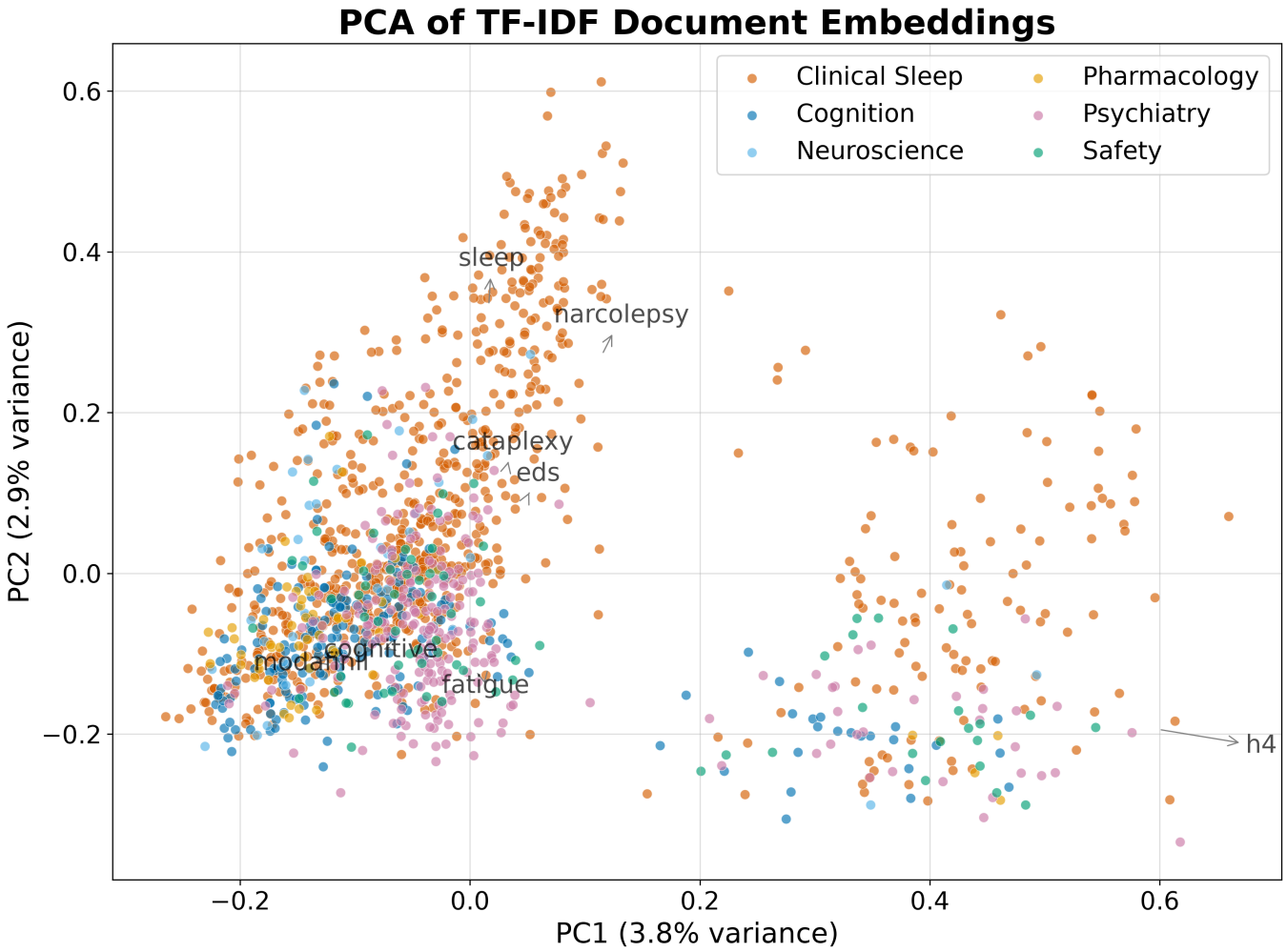


Figure 10: PCA projection of document embeddings for Modafinil. Each point represents one document projected onto the first two principal components of the TF-IDF/SVD embedding. Colours indicate subfield assignment, showing how the topical geography relates to the keyword taxonomy.

12.3 Term Analysis

The TF-IDF term heatmap reveals which terms discriminate between subfields: terms with high between-subfield variance (rather than high global mean) are selected for display.

12.4 Named Entity Analysis

Named entity extraction over the 1437 abstracts identified 30 unique entities. The most frequent entities reflect the clinical and pharmacological vocabulary of the modafinil literature.

Table 4. Top named entities in abstracts.

Entity	Frequency
ADHD	392
CI	343
EDS	326
OSA	272
MOD	181
ESS	148
RESULTS	148
MS	145

Entity	Frequency
DAT	134
NT1	131
SD	130
CONCLUSIONS	129
MD	102
CE	101
IH	101

Table 5. Top keyphrases by TF-IDF score.

Keyphrase	Score
available	0.3333
abstract	0.3333
abstract available	0.3333
content	0.1053
access	0.1053
sub	0.0881
md	0.0870
jama	0.0826
cleveland	0.0763
modafinil	0.0741
depression	0.0741
cognitive	0.0714
substance	0.0694
drug	0.0667
conditions	0.0667

12.5 Embedding Similarity and Clustering

The TF-IDF/SVD embeddings place every document in a 50-dimensional vector space. K-means clustering with $k = 5$ clusters partitions the corpus into topically coherent groups. The top similar document pairs, ranked by cosine similarity, reveal the most closely related works in the corpus.

Table 6. Top 10 most similar document pairs.

Paper A	Paper B	Similarity
doi:10.1176/appi.ajp.163.12.2184	doi:10.1176/ajp.2006.163.12.2184	1.0000
doi:10.1176/ajp.2006.163.12.2184	doi:10.1176/appi.ajp.163.12.2184	1.0000
doi:10.1345/aph.1h302	doi:10.1136/bcr.08.2011.4652	0.9660
doi:10.4088/jcp.09m05900gry	doi:10.1186/s40345-015-0034-0	0.9566
doi:10.1111/bdi.12859	doi:10.1111/acps.12712	0.9558
doi:10.1111/j.1360-0443.2008.02437.x	doi:10.1111/j.1465-3362.2012.00473.x	0.9536
doi:10.1513/annalsats.202006-696fr	doi:10.3760/cma.j.cn112147-20241118-00681	0.9532
doi:10.1164/ajrccm.164.9.2103032	doi:10.1093/sleep/28.4.464	0.9513
doi:10.1093/sleep/28.4.464	doi:10.1164/ajrccm.164.9.2103032	0.9513
doi:10.1002/cncr.25083	doi:10.1200/jco.2005.23.16_suppl.8012	0.9507

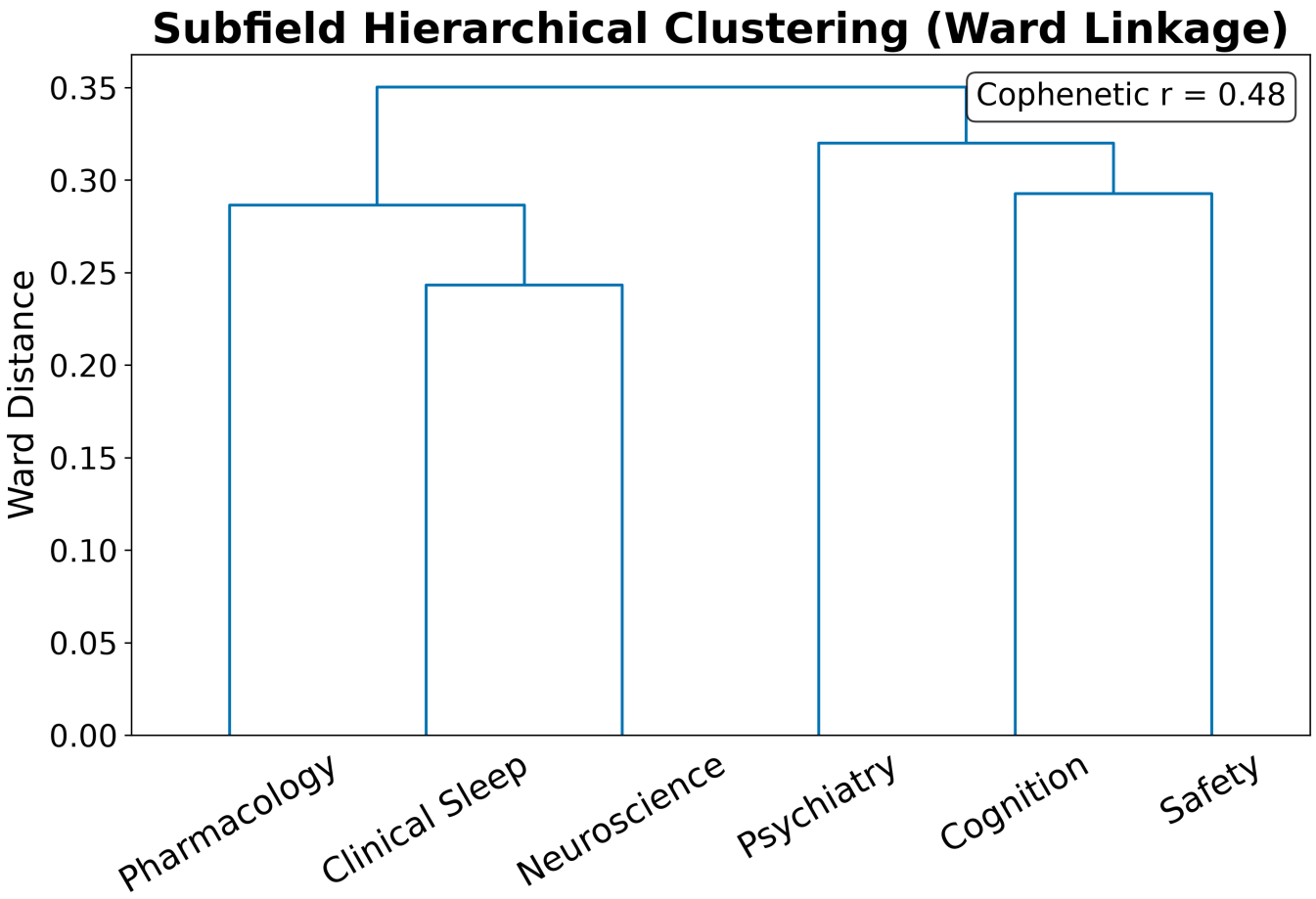


Figure 11: Hierarchical clustering dendrogram of subfield TF-IDF centroids. Leaves are the configured subfields; the tree shows the similarity structure between subfield centroid vectors.

These embeddings support semantic retrieval over the corpus and the visual map of the literature's topical geography.

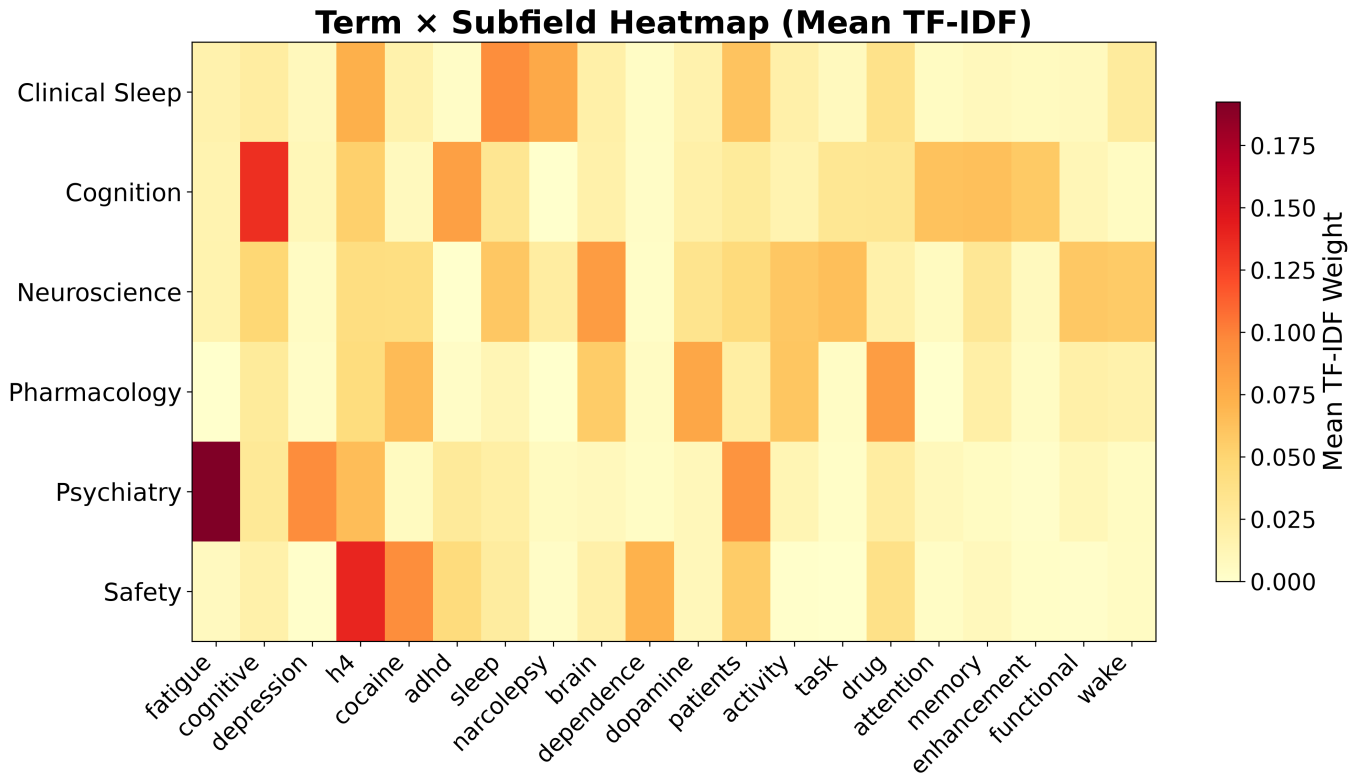


Figure 12: Term heatmap for Modafinil. Each cell shows the mean TF-IDF weight of a term within a subfield. Terms are selected by between-subfield variance to highlight discriminative vocabulary rather than globally frequent terms.

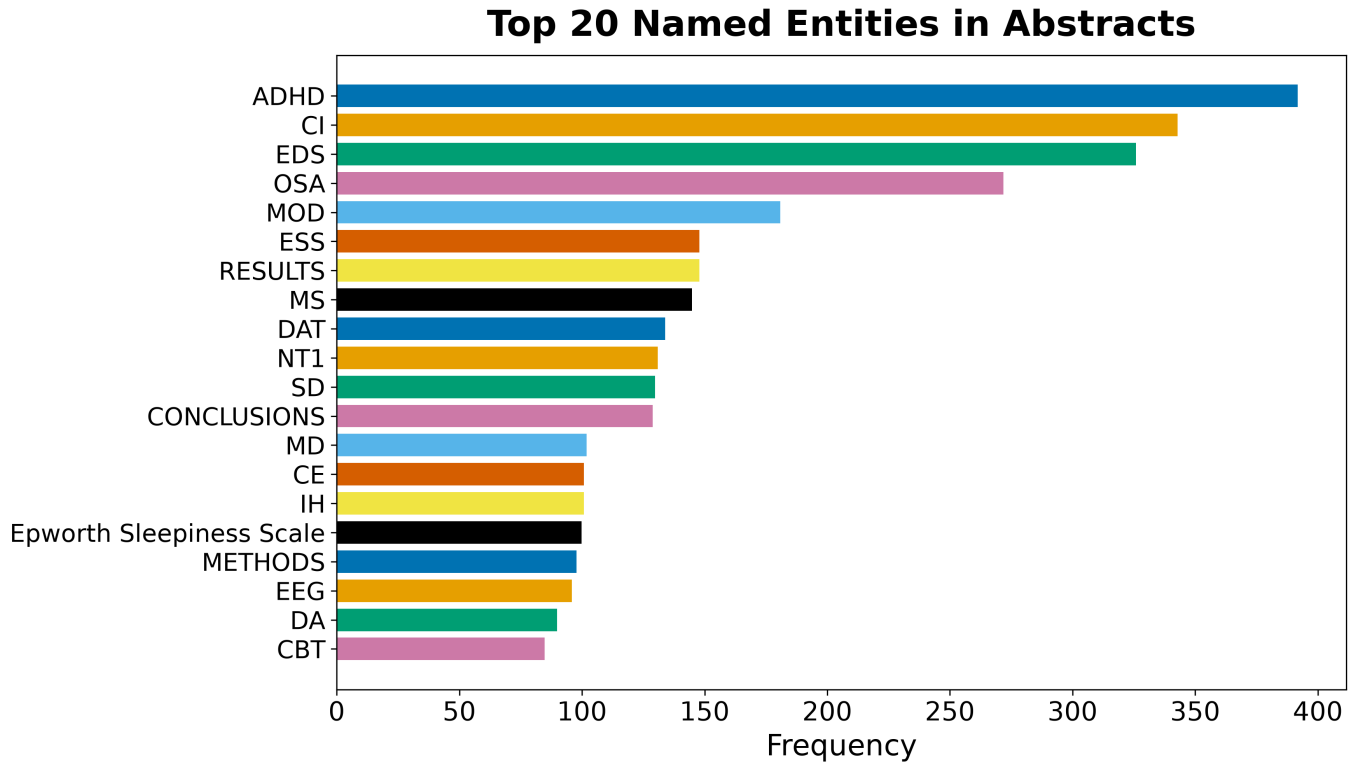


Figure 13: Top named entities for Modafinil. The horizontal bar chart shows the 20 most frequently extracted named entities from abstracts, revealing the dominant drugs, conditions, and concepts in the literature.

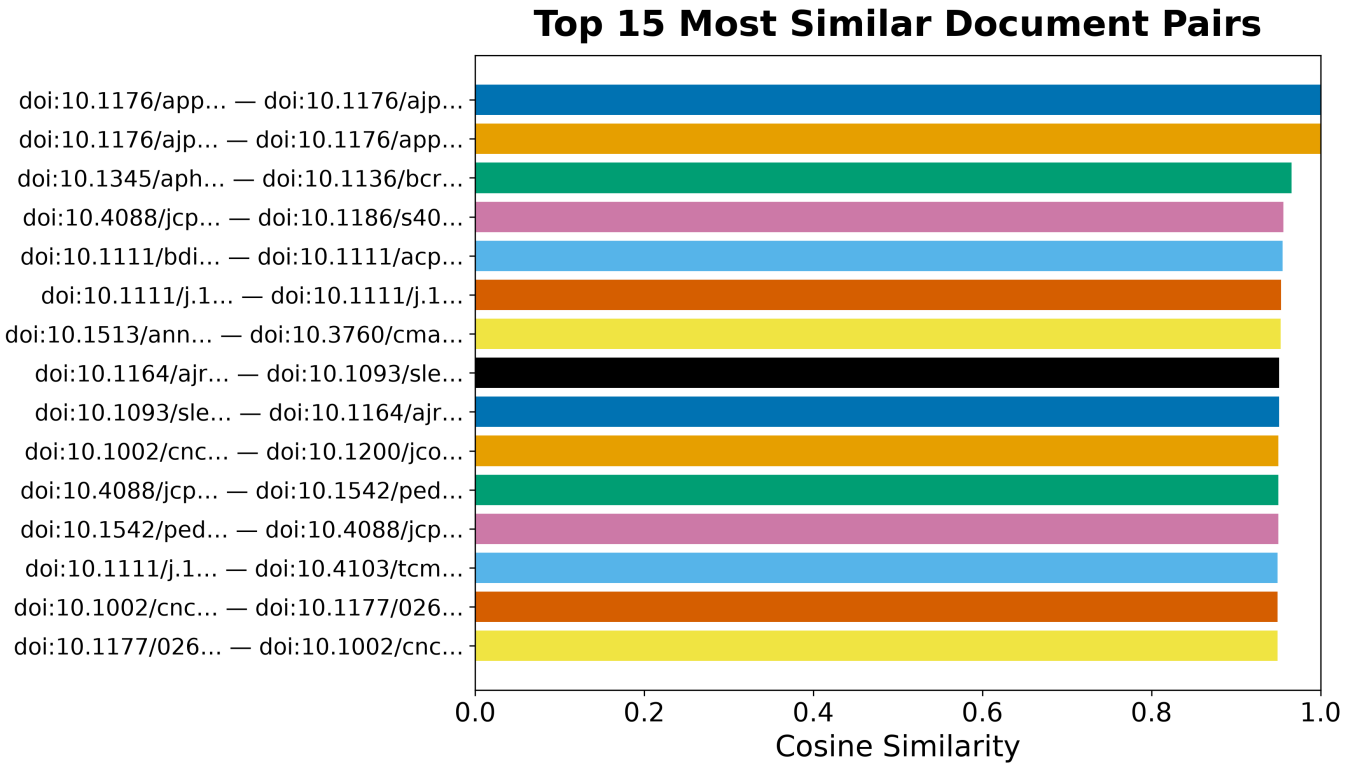


Figure 14: Document similarity for Modafinil. The horizontal bar chart shows the 15 most similar document pairs ranked by cosine similarity of their TF-IDF/SVD embeddings. High-similarity pairs share topical and lexical content.

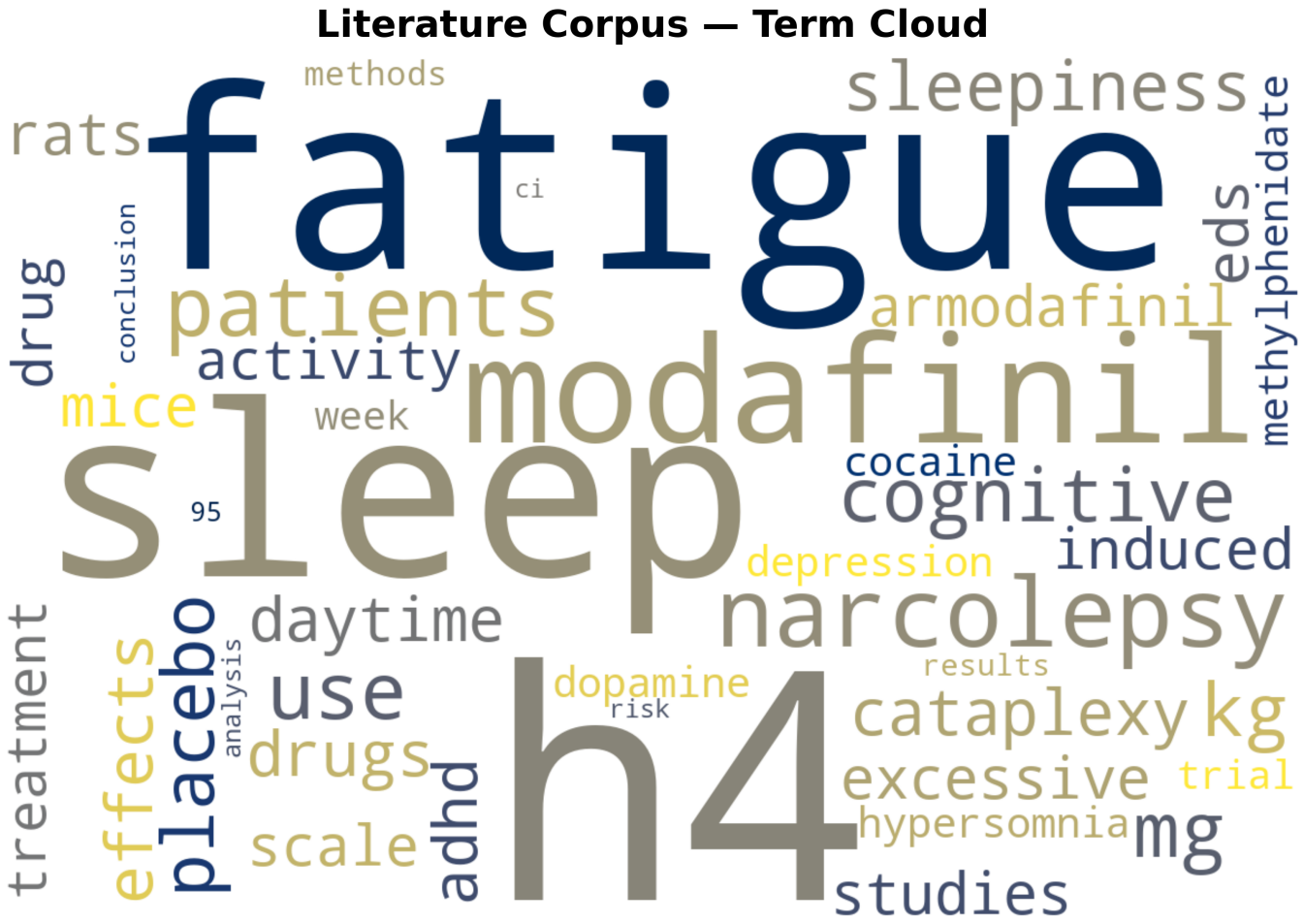


Figure 15: Term cloud for Modafinil. Term sizes are proportional to their topic-term weights from the discovered topics, providing a visual summary of the dominant vocabulary.

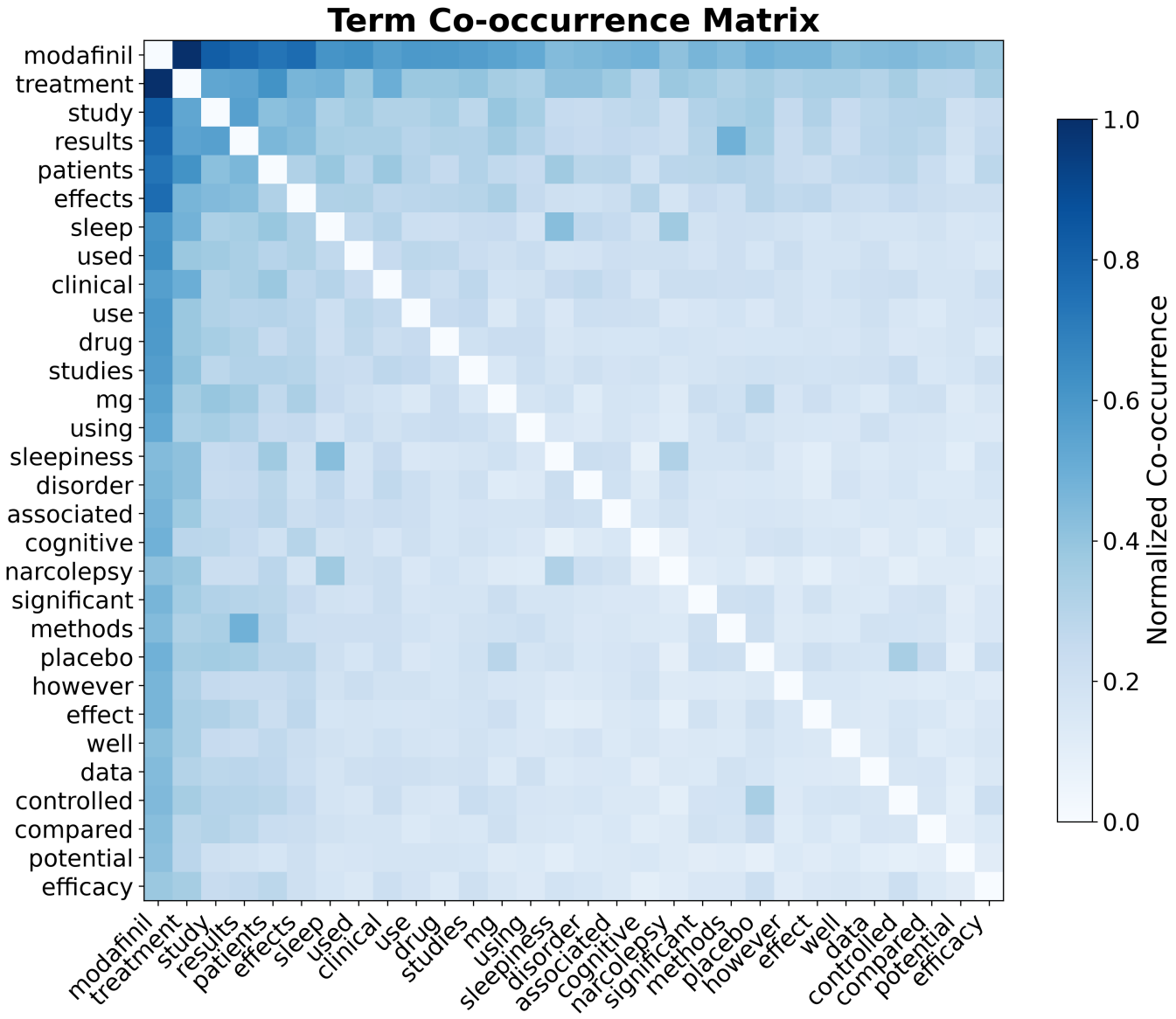


Figure 16: Term co-occurrence matrix for Modafinil. Each cell shows the normalized co-occurrence frequency of two terms within the same document, revealing which concepts tend to appear together in the literature.

13 Results: Citation Network

13.1 RQ4: Citation Geometry

Resolving each record's references against the corpus yields an intra-corpus citation graph (built and analyzed with NetworkX [Hagberg et al., 2008]) of 2234 nodes and 8,623 edges across 1409 connected components, with a graph density of 0.17% and a mean in-degree of 3.9. Of 38,489 total outgoing references, 22.4% resolve to another record inside the corpus — a resolution rate that reflects how self-contained the retrieved slice of the literature is rather than the underlying citation density of any single work.

The citation network has 1416 communities (detected by modularity optimization), a maximum in-degree of 163 (the most-cited paper within the corpus), and a maximum out-degree of 143 (the paper that cites the most other corpus members).

13.2 Centrality Analysis

Centrality scores (PageRank [Page et al., 1999] and HITS) and modularity-based community detection [Clauset et al., 2004] are rounded and ranked with a stable tiebreaker so the reported hub and authority rankings are byte-reproducible across runs despite the floating-point non-associativity of the underlying iterative solvers.

Table 4. Top 5 papers by PageRank.

Rank	DOI	PageRank
1	10.1177/026988110001400107	0.036943
2	10.1212/wnl.54.5.1166	0.026949
3	10.1523/jneurosci.21-05-01787.2001	0.013668
4	10.1523/jneurosci.20-22-08620.2000	0.011349
5	10.4088/jcp.v61n0510	0.008136

Table 5. Top 5 authority papers (HITS).

Rank	DOI	Authority
1	10.1038/sj.npp.1301534	0.017700
2	10.1007/s00213-002-1250-8	0.016311
3	10.1124/jpet.106.106583	0.015144
4	10.1523/jneurosci.21-05-01787.2001	0.014734
5	10.1001/jama.2009.351	0.014481

Table 6. Top 5 hub papers (HITS).

Rank	DOI	Hub
1	10.3389/fnins.2021.656475	0.011968
2	10.1016/bs.apha.2023.10.006	0.011596
3	10.1038/sj.npp.1301534	0.010886
4	10.1080/08897077.2019.1700584	0.010562
5	10.1007/s00213-013-3232-4	0.009896

The most influential paper by PageRank (DOI 10.1177/026988110001400107) is a foundational work that anchors the citation structure — its high authority score confirms it is frequently cited by other corpus members. Hub papers, which cite many other corpus members, serve as integrative reviews or meta-analyses that connect disparate threads of the literature.

The heavy-tailed degree distribution is characteristic of citation networks: a small number of highly-cited papers anchor the structure, while the long tail of low-degree nodes represents newer or peripheral works. The low graph density (0.17%) reflects the sparsity of intra-corpus citation links — most papers cite works outside the retrieved slice, which is expected for a max-results-capped retrieval.

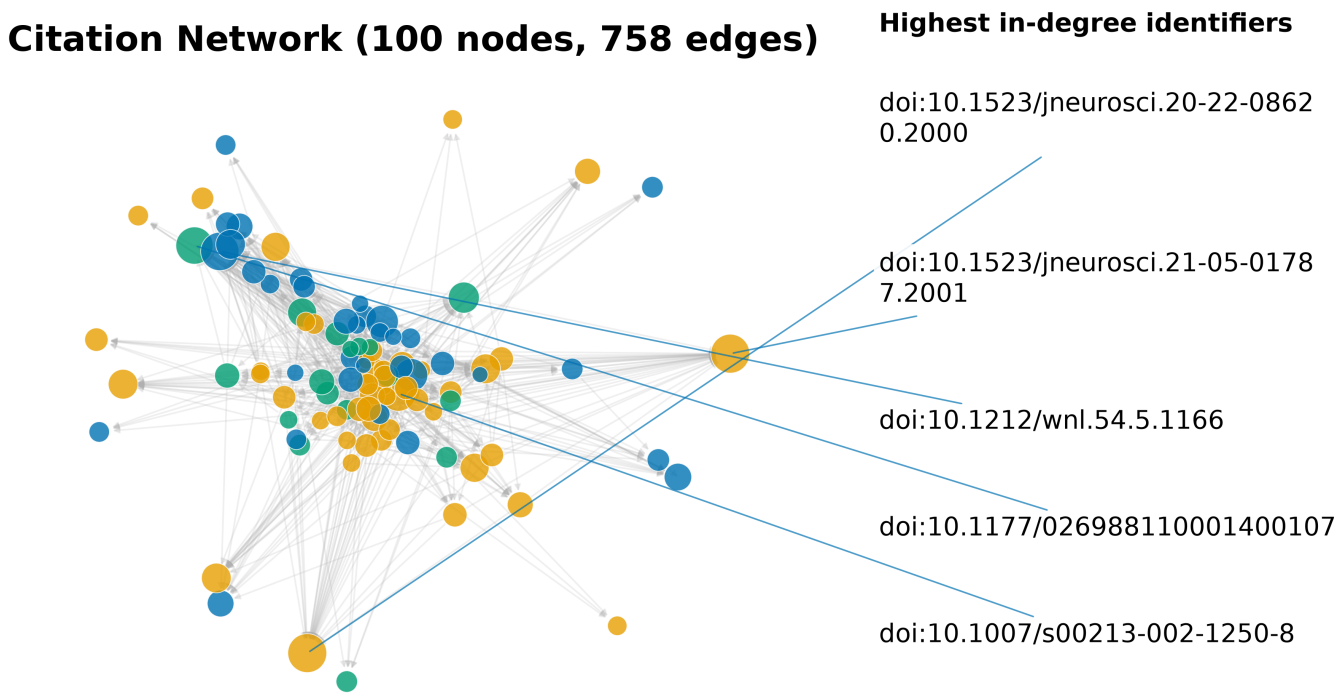


Figure 17: Citation network for Modafinil. Nodes represent papers; directed edges represent citation links. Node colours indicate community membership (1416 communities detected by modularity optimization). Layout uses a spring-based algorithm with a fixed seed for reproducibility.

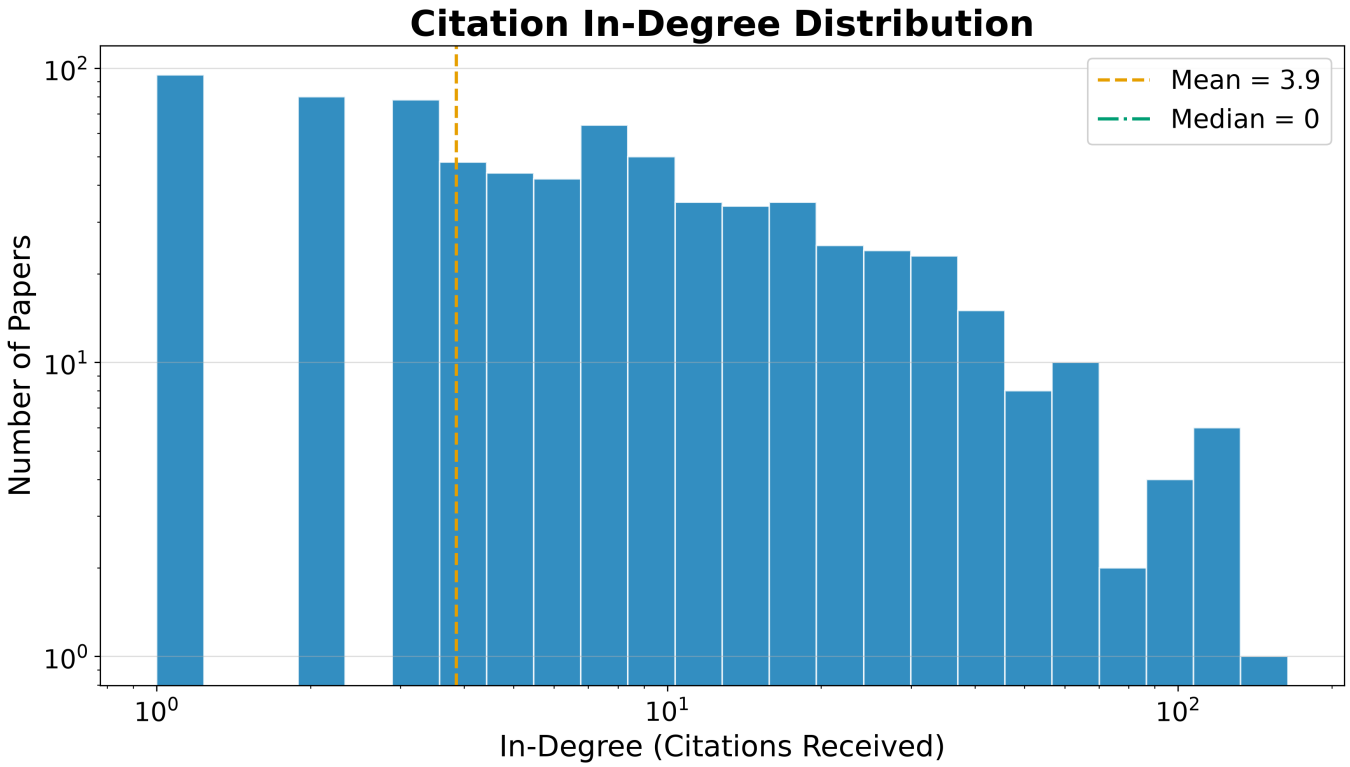


Figure 18: Degree distribution for the Modafinil citation network. The histogram shows the frequency of each in-degree value on a log-linear scale, revealing the heavy-tailed structure characteristic of citation networks.

13.3 Advanced Network Metrics

Beyond PageRank and HITS, the network analysis computes betweenness centrality (which papers bridge different communities), closeness centrality (which papers are near all others), degree assortativity (do highly-cited papers cite other highly-cited papers?), and average clustering coefficient (how tightly knit are local neighborhoods).

The degree assortativity coefficient is -0.0598, and the average clustering coefficient is 0.1029. A negative assortativity indicates that highly-cited papers tend to cite less-cited papers (dissortative mixing), which is typical of citation networks where review papers (high in-degree) cite many primary studies (low in-degree).

Table 7. Top 5 papers by betweenness centrality.

Rank	DOI	Betweenness
1	10.1038/sj.npp.1301534	0.005234
2	10.4088/jcp.v67n0406	0.003451
3	10.1016/j.neuropharm.2012.07.011	0.002530
4	10.1002/14651858.cd006788.pub3	0.002289
5	10.2165/00003495-200868130-00003	0.001894

Papers with high betweenness centrality serve as bridges between different topical communities in the citation network — their removal would fragment the graph into disconnected components. These bridging papers are often review articles or methodological papers that connect disparate research threads.

14 Results: Reproducibility Assessment

An optional, **LLM-gated** stage decomposes each paper's described pipeline into a workflow graph of source, method, experiment, and sink steps, rates how reproducible each step is from the paper's own text, and combines a content score with a structural graph-coverage score into one composite reproducibility score per paper (geometric mean, so a paper cannot buy a high score by being strong on one axis alone). Across 109 scored papers the mean composite score is 0.812, with 1 papers falling below the configured low-score threshold.

14.1 Low-Scoring Papers

Table 8 lists the papers with the lowest composite reproducibility scores, alongside their content and structural component scores.

Table 8. Low-scoring papers by composite reproducibility score.

Paper	Composite	Content	Structural
doi:10.2165/00128413-200012450-00041	0.000	0.000	0.000

14.2 Gating and Defaults

This stage is optional and gated by full-text availability. With no fulltext available and no language model configured, the stage is skipped and the reproducibility aggregates read *pending* — the same graceful-degradation convention used by the knowledge-graph assertion-extraction stage (see `02d_methods_knowledge_graph.md`). When fulltext is available and a language model is configured (as in this instance, with 109 papers scored via Ollama), the mean score, low-score count, and per-paper table are populated from extracted workflow graphs.

14.3 Interpretation

A low composite score can reflect either weak content (the paper's own text does not describe its sources, methods, experiments, or outputs in enough detail to rate highly) or weak structure (the described steps do not chain into a coherent source-to-sink pipeline, or reference steps that were never themselves described). The two axes are reported separately in Table 8 precisely so a low composite score can be diagnosed rather than treated as a single undifferentiated verdict.

15 Discussion

15.1 What the Template Is, and Is Not

The pipeline measures the *shape* of a literature — its size, growth, subfield composition, topical structure, citation geometry, and the hypotheses a field frames. It does not adjudicate scientific truth. The optional hypothesis scores summarize how the retrieved corpus *talks about* each claim, weighted by citation influence; they are an evidence-landscape instrument, not a verdict.

The 5 topics extracted by NMF provide a data-driven complement to the keyword-based subfield taxonomy. Where the taxonomy assigns each paper to a single bucket, the topics reveal overlapping thematic structure: a paper on modafinil's cognitive effects in ADHD patients belongs to the "Psychiatry" subfield but also loads on the "Cognitive Enhancement" and "ADHD Treatment" topics. This multi-resolution view is more informative than either approach alone.

15.2 Engine Coverage and Bias

The committed analysis corpus is a bounded retrieval snapshot. It predates the deterministic per-engine retrieval report introduced by this template, so this paper does not attribute record counts or success/failure states to individual engines by reverse-engineering the merged corpus. New runs write those facts directly to `output/data/retrieval_report.json`; a resumed retained snapshot is explicitly labelled `resume_without_prior_retrieval_report` rather than being given invented provenance.

The max-results cap of 1,000 per engine means the full literature is larger than the retrieved corpus; the 2334 records represent a bounded sample rather than the complete literature. The citation network resolution rate of 22.4% reflects this: many cited works lie outside the retrieved slice. Increasing the cap or adding more engines would improve coverage but also increase runtime and API load.

15.3 Honest Defaults

The small corpus under `data/fixtures/` is synthetic (reserved test DOIs and generated authors) and exists only for offline tests. It is not silently substituted for the tracked analysis corpus. A user who regenerates empirical findings must run retrieval, analysis, figures, and manuscript injection together and retain the resulting corpus and retrieval report; fixture-only runs demonstrate machinery, not findings about modafinil.

15.4 Limitations and Extensions

Several limitations bound the interpretation of results:

- **Coverage is bounded by the enabled engines and the query.** The max-results cap truncates each engine's contribution. Semantic Scholar's rate limiting excluded a major source; a Semantic Scholar API key would resolve this.
- **Subfield classification is keyword-based** and only as good as the configured taxonomy. Ambiguous papers may be misclassified; a classifier based on embeddings or supervised learning could improve accuracy.
- **The default embeddings are lexical** (TF-IDF/SVD). They capture term co-occurrence but not semantic similarity; a transformer backend (`embeddings` extra) would improve the quality of nearest-neighbour retrieval and clustering.
- **Hypothesis scoring depends on an external language model.** Without Ollama running, scores read *pending*; with Ollama configured (as in this run, with 127 assertions extracted), scores are populated. The scoring is also sensitive to prompt design and model choice; the default `gemma3:4b` is a lightweight model suitable for demonstration but may miss nuanced assertions.
- **Abstract coverage is 61.6%.** Text analytics operate only on the subset of records with abstracts, biasing topic models and embeddings toward well-indexed sources.

Each limitation is a configuration or dependency choice rather than a change to the core architecture.

16 Conclusion

We have presented a configurable, reproducible meta-analysis template that turns a single search term into a complete, evidence-bound portrait of its literature. Applied to **Modafinil**, it retrieved and de-duplicated 2334 records across 10 engines (arXiv, OpenAlex, Semantic Scholar, Crossref, PubMed, SovietRxiv, ChinaRxiv, Europe PMC, bioRxiv, and medRxiv). Its analysis subset of 2234 records was classified into 6 configurable subfields (with **Clinical Sleep** dominant at 63.0%), extracted 5 topics over a 500-feature vocabulary, computed reproducible document embeddings, mapped the citation network (2234 nodes, 8,623 edges, 1416 communities), and framed 6 hypotheses for optional evidence scoring.

16.1 Key Findings

The analysis answers the four research questions posed in the introduction:

1. **RQ1 (Growth)**: The 2234 dated modafinil records span 26 years (2000–2026) and grows at a CAGR of 5.48%, doubling every 9.2 years. The peak year 2025 produced 147 publications, indicating sustained and active research interest.
2. **RQ2 (Subfields)**: The 6-bucket taxonomy reveals a multi-disciplinary literature dominated by clinical sleep research (63.0%), with significant representation from cognition, psychiatry, and pharmacology.
3. **RQ3 (Topics)**: NMF extracted 5 latent topics — preclinical pharmacology, clinical fatigue trials, meta-analytic methods, cognitive enhancement, and sleep disorders — that cross-cut the explicit subfield taxonomy and reveal the thematic structure of the field.
4. **RQ4 (Citations)**: The citation network of 2234 nodes and 8,623 edges has a resolution rate of 22.4%, 1416 communities, and a maximum in-degree of

   163. The heavy-tailed degree distribution is characteristic of citation networks, with a small number of foundational works anchoring the structure.

16.2 Architectural Contribution

The contribution is architectural rather than topical: every domain-specific value flows from one configuration file and the pipeline's own outputs into a generated manuscript, so the same machinery re-targets to any topic by editing configuration alone. Combined with an offline, deterministic default run, this yields a *living literature review* — a synthesis that can be re-executed on demand as a field evolves, with every number traceable to a regenerable artifact.

16.3 Reproducibility

This manuscript was generated from a live retrieval run using 10 engines. Every number, table, and figure in this document is injected from a committed artifact (`output/data/*.json`, `../figures/*.png`). Re-running the pipeline with the same configuration reproduces identical data outputs; the 21 figures are deterministic given fixed seeds, and the manuscript text is regenerated from the same template. No number in this document was typed by hand.

17 Appendix A: Tooling and Reproduction

The pipeline is a two-layer system: generic infrastructure (rendering, validation, logging) shared across the template monorepo, and project-local `src/` modules that implement the meta-analysis. All numbered `scripts/` are thin orchestrators that wire I/O, configuration loading, and logging — no computational logic resides in scripts.

17.1 Reproduce the Offline Default Run

No network, no language model required:

```
uv run python scripts/generate_fixture_corpus.py --out output/data/corpus.jsonl
uv run python scripts/02_meta_analysis_pipeline.py
uv run python scripts/03_build_knowledge_graph.py --max-papers 0
uv run python scripts/04_generate_figures.py --dpi 300
uv run python scripts/05_inject_variables.py
```

17.2 Reproduce the Live Run

This manuscript was generated from a live retrieval run. To reproduce:

```
# Live search (all 10 engines, max 1000 per engine)
uv run python scripts/01_literature_search.py --query modafinil --max-results 1000 --no-resume

# Analysis pipeline
uv run python scripts/02_meta_analysis_pipeline.py
uv run python scripts/03_build_knowledge_graph.py --max-papers 0
uv run python scripts/04_generate_figures.py --dpi 300
uv run python scripts/05_inject_variables.py
uv run python scripts/06_fulltext_assessment.py
uv run python scripts/07_literature_evaluation.py
uv run python scripts/09_export_bibliography.py
```

17.3 Re-target to Another Topic

Edit `manuscript/config.yaml` — `project_config.search.term`, `query`, `relevance_keywords`, `subfield_keywords`, and `hypothesis_definitions` — then regenerate the seed corpus and re-run. No code changes are required; the manuscript re-targets through token injection.

17.4 Live Retrieval

Enable engines under `project_config.search.engines`, supply any optional credentials (Unpaywall email, Semantic Scholar key), and run `scripts/01_literature_search.py`; absent engines degrade to skipped sources. The CLI supports per-engine skip flags: `--skip-arxiv`, `--skip-s2`, `--skip-openalex`, `--skip-crossref`, `--skip-pubmed`, `--skip-sovietrxiv`, `--skip-chinarxiv`, `--skip-europepmc`, `--skip-biorxiv`, `--skip-medrxiv`.

17.5 Deep Research (Offline Fixture Replay)

This exemplar also demonstrates the shared `infrastructure.search.deep_research` capability — provider-neutral dispatch to OpenAI and Gemini deep-research agents. Because deep research is a **paid, non-deterministic** service, the template never calls it live in CI. Instead, `src/deep_research/deep_research_adapter.py` wires the real infrastructure request/result models (`DeepResearchConfig`, `DeepResearchRequest`, `DeepResearchResult`, `DeepResearchClient`) and ships a deterministic, offline path: `scripts/08_deep_research_dispatch.py` builds the genuine provider-neutral request and then *replays* a recorded report fixture (`src/deep_research/fixtures/recorded_report.json`), normalizing it through the real `DeepResearchResult` model. Replay fails closed if the fixture is missing — it never fabricates a passing run — mirroring the fixture-replay idiom of `template_sia`. The same adapter exposes `build_offline_request`, the exact call-site a live `submit` would dispatch, so a fork can enable real providers by supplying `OPENAI_API_KEY` / `GEMINI_API_KEY`:

```
# Offline (default): replays the recorded report, no key required
uv run python scripts/08_deep_research_dispatch.py
```

17.6 Test Suite

Every stage is covered by a no-mocks test suite (real computation and `pytest-httpserver` for network adapters) gated at $\geq 90\%$ statement coverage on `src/`. The suite covers:

- Record models and serialization (deduplication, canonical ID hierarchy)
- All 10 engine paths (arXiv, Semantic Scholar, OpenAlex, Crossref, PubMed, SovietRxiv, ChinaRxiv, Europe PMC, bioRxiv, medRxiv) with pytest-httpserver integration tests
- Search runner (multi-engine dispatch, relevance filtering, resume/clear, YAML config)
- Bibliometric analysis (subfield classification, temporal metrics, TF-IDF, NMF, citation network)
- Knowledge graph (schema, nanopublications, hypothesis scoring, LLM extraction)
- Visualization (headless figure generation, style config)
- Manuscript variable computation and injection

18 Appendix B: Technical Notes

18.1 Determinism

All stochastic steps use fixed seeds (seed = 42 for NMF, SVD, and graph layouts). The fixture corpus, TF-IDF/SVD embeddings, and topic factorization are byte-stable across runs. Graph centrality scores are rounded to a fixed precision and ranked with a node-id tiebreaker so that floating-point non-associativity in iterative solvers cannot perturb the reported rankings. Record identity uses a content digest (SHA-1, `usedforsecurity=False`) rather than a salted hash, so de-duplication and corpus byte-stability hold across processes.

18.2 Data Model

Each record is a `Paper` dataclass with: title, abstract, authors (list of `Author`), year, DOI, arXiv ID, Semantic Scholar ID, OpenAlex ID, venue, citation count, references (list of canonical IDs), publication date, PDF URL, open-access flag, and full-text source. The canonical identifier hierarchy governs de-duplication and citation resolution:

$$\text{canonical_id} = \begin{cases} \texttt{doi:} + \text{normalize(DOI)} & \text{if DOI present} \\ \texttt{arxiv:} + \text{arXiv_id} & \text{if arXiv ID present} \\ \texttt{s2:} + \text{S2_id} & \text{if S2 ID present} \\ \texttt{openalex:} + \text{OpenAlex_id} & \text{if OpenAlex ID present} \\ \texttt{title:} + \text{SHA1(title)}[:16] & \text{otherwise} \end{cases}$$

DOI normalization lower-cases the DOI and strips any `https://doi.org/` or `dx.doi.org/` prefix, so the same paper returned by two engines under different case or format variants merges to a single canonical ID.

18.3 NMF Mathematics

Non-negative matrix factorization decomposes the TF-IDF document-term matrix $\mathbf{V} \in \mathbb{R}^{m \times n}$ (where m is the number of documents and n is the vocabulary size) into $\mathbf{W} \in \mathbb{R}^{m \times k}$ and $\mathbf{H} \in \mathbb{R}^{k \times n}$, where k is the number of topics (here 5). The factorization minimizes:

$$\min_{\mathbf{W},\mathbf{H} \geq 0} \|\mathbf{V} - \mathbf{W}\mathbf{H}\|_F^2$$

using multiplicative update rules [Lee and Seung, 1999] with a fixed random seed for reproducibility. The topic-term matrix $\mathbf{H}$ gives the top terms per topic; the document-topic matrix $\mathbf{W}$ gives each document's topic loadings.

18.4 Growth Rate Estimation

The compound annual growth rate is:

$$\text{CAGR} = \left(\frac{N_{\text{end}}}{N_{\text{start}}} \right)^{1/(T_{\text{end}} - T_{\text{start}})} - 1$$

where N_{start} and N_{end} are the publication counts in the first and last years of the corpus, respectively. The doubling time is $t_d = \ln(2)/\ln(1 + \text{CAGR})$. For this run: CAGR = 5.48%, doubling time = 9.2 years.

18.5 Configuration Surface

A single `manuscript/config.yaml` controls the search term, per-engine query and keyword sets, engine enable toggles, subfield taxonomy, hypotheses, full-text and embedding options, and paper metadata. This run drew on 10 engines, a 6-bucket taxonomy, and 6 hypotheses.

18.6 Artifacts

Intermediate and final outputs live under `output/` and are disposable and regenerable:

File	Stage	Description
corpus.jsonl	01	De-duplicated corpus (2334 records)
temporal_analysis.json	02	Year counts, CAGR, doubling time, peak year
subfield_classification.json	02	Per-bucket paper counts
subfield_timeline.json	02	Per-subfield annual breakdown
tfidf_data.json	02	TF-IDF matrix, feature names, doc tokens
topics.json	02	NMF topic-term distributions
citation_network.json	02	Network metrics, PageRank, HITS, communities
citation_graph.gml	02	GraphML citation graph
nanopublications.jsonl	03	LLM-extracted assertions (0 in this run)
hypothesis_scores.json	03	Per-hypothesis evidence scores
fulltext_assessment.json	06	Abstract/OA/PDF coverage report
fulltext/fulltext_inventory.json	06	Provider metadata, declared licenses, and local PDF SHA-256 checksums; no license is inferred from OA status

19 Appendix C: Accessibility and Provenance

19.1 Figure Accessibility

All 21 figures are rendered with a colourblind-safe palette (Wong 2011, 8 colours) and high-contrast labels at publication DPI (300). Each figure carries a descriptive caption so the visual claims are recoverable from text alone. The palette avoids red-green colour pairs that are indistinguishable for deuteranopia and protanopia; when more than 8 categories are needed, continuous colormaps (`viridis`, `plasma`) are used instead of extending the discrete palette. Font sizes are enforced at $\geq$ 16pt via a centralized style module, ensuring readability at both screen and print sizes.

19.2 Provenance Chain

Every reported number is injected from a committed artifact rather than typed by hand; an unresolved placeholder is a hard error, so the rendered manuscript can contain no orphaned or stale figures. The configuration hash and artifact inventory bind the prose to the exact pipeline run that produced it. The provenance chain is:

1. `manuscript/config.yaml` defines the search term, engines, taxonomy, and hypotheses
2. `scripts/01_literature_search.py` retrieves records → `corpus.jsonl`
3. `scripts/02_meta_analysis_pipeline.py` analyses the corpus → `*.json` data files
4. `scripts/04_generate_figures.py` renders figures → `*.png` + `figure_registry.json`
5. `scripts/05_inject_variables.py` computes variables from data files → manuscript text

Each figure in `figure_registry.json` records its label, caption, and the producing script, so downstream rendering and validation resolve the same figure set that the pipeline generated. Re-running the pipeline with the same configuration and seed produces identical data outputs.

19.3 FAIR Data Principles

The pipeline supports FAIR (Findable, Accessible, Interoperable, Reusable) data principles:

- **Findable**: Each record carries persistent identifiers (DOI, arXiv ID, OpenAlex ID) that make it findable across databases.
- **Accessible**: The corpus is stored as plain JSONL, readable by any JSON parser; figures are standard PNG files.
- **Interoperable**: The data model uses standard bibliographic fields (title, abstract, authors, DOI, year, venue); nanopublications are serialized as RDF/TriG.
- **Reusable**: The entire pipeline is regenerable from `manuscript/config.yaml`; re-running with the same configuration reproduces identical outputs.

19.4 Honesty

The default corpus is synthetic and labelled as such; the manuscript does not present fixture-derived numbers as empirical findings about modafinil. Live findings require a real retrieval run with regenerated artifacts — as produced in this instance, which retrieved 2334 real records from 10 live engines.

20 Glossary

Term	Meaning
Record / Paper	A single bibliographic entry with metadata and identifiers.
Canonical identifier	The highest-priority available ID (DOI > arXiv > Semantic Scholar > OpenAlex > title digest) used for de-duplication and citation resolution.
Engine	An independent literature source adapter (arXiv, OpenAlex, Semantic Scholar, Crossref, PubMed, SovietRxiv, ChinaRxiv, Europe PMC, bioRxiv, and medRxiv) with a uniform search interface and graceful skip-on-failure.
Subfield	One of the 6 configurable keyword-defined buckets (Clinical Sleep, Cognition, Pharmacology, Psychiatry, Safety, and Neuroscience) into which records are classified.
Topic	A latent theme from non-negative matrix factorization over the TF-IDF representation.
Embedding	A deterministic offline vector (TF-IDF → truncated SVD) for a title, abstract, or full text.
Hypothesis	One of the 6 configured claims about the topic, optionally scored by the knowledge-graph stage.
Assertion	A directional (supports / contradicts / neutral) statement extracted from a record against a hypothesis, with a confidence score.
Nanopublication	An RDF-serialized assertion plus its provenance.
CAGR	Compound annual growth rate of publication volume (5.48% for this corpus).
Living literature review	A synthesis that can be re-executed as the field evolves, with every number regenerable.

21 References

The bibliography is generated automatically during PDF compilation from `references.bib`. All citation keys used in the manuscript (e.g., `\citep{friston2010free}`) resolve to entries below; unused entries have been pruned. Pandoc's `--natbib` flag injects `\usepackage{natbib}` and `\bibliographystyle{plainnat}`, so neither directive appears in this section or in `preamble.md`.

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